ATARNotes

Physics Units 3&4

Edition 2 (2017-2021)

COMPLETE COURSE NOTES

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- Written in a way that's easy to understand, so you can immediately grasp key concepts.
- Graduating with a 99.50 and a 49 raw in Physics, Alwin has since delivered Physics lectures and tutorials to thousands of students across the state.

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Alwin Wang

ATAR Notes

VCE Physics Units 3&4
Complete Course Notes
Edition 2 (2017-2021)

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Preface

Hey everyone, welcome to Physics Units 3&4!

I'm Alwin, and in these notes, I'll be trying my best to help you throughout the year. The 2017–2021 study design has some big changes, so I'll provide you with all the resources I have, including everything from past textbooks and my own notes for university-level Physics courses. These notes aren't supposed to overwhelm you in any way, so I've tried to flag the especially important concepts and where relevant.

Unlike previous study design alterations, the change from 2016 to 2017 for Units 3&4 is a large one. While you can find a complete document of the new study design on the VCAA website, here is a quick, rough guide of what's changed in the new study design so that you know what to look for when reading through past exams or other resources:

- There has been a reshuffle of topics in the old study design into new Area of Studies.
- The detailed studies section of the course has now been removed with many parts incorporated into the main study design.
- Pre-2017 *Unit 3: Motion in one and two Dimensions* is still examined, but it is now split up into different parts in the new study design in Unit 3.
- Pre-2017 *Unit 3: Electronics and Photonics* is now mostly in Unit 1.
- Pre-2017 *Unit 4: Electric Power* has now been extended with a larger focus on fields in Unit 3.
- Pre-2017 *Unit 4: Interactions of Light and Matter* has now been significantly extended with a more detail on the wave and particle models for light in Unit 4.
- Pre-2017 *Detailed Study: Einstein's Relativity* has now been almost entirely been incorporated into Unit 3.
- Pre-2017 *Detailed Study: Materials and their use in Structures* is now one of twelve options for Area of Study 2 in Unit 2.
- Pre-2017 *Detailed Study: Further Electronics* is now one of twelve options for Area of Study 2 in Unit 2.
- Pre-2017 *Detailed Study: Synchrotron and its Applications* is now one of twelve options for Area of Study 2 in Unit 2 with a small section about accelerating charges in a field moved into Unit 3.
- Pre-2017 *Detailed Study: Photonics* has now been moved into Unit 4, although it has a large section on fibre optic cables which is not specifically asked for in the new study design.
- Pre-2017 *Detailed Study: Sound* has now been moved into Unit 4, although it has a large section on microphones and speakers which is not asked for in the new study design.

Again, this is just a very rough list but it does cover most of the new study design. Hopefully it should help you with finding resources!

I really encourage you to use your textbook, study notes and of course pay attention in class to your teacher in conjunction with these notes to get the most out of Physics this year.

All the best,

— Alwin Wang

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Part I

Unit 3: How do fields explain motion and electricity?

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Area of Study 1

How do things move without contact?

1.1 Fields and Interactions

1.1.1 The Field Model

In VCE Physics we deal with what's known as classical fields. These are gravitational fields and electromagnetic fields. A field is the assignment of a physical quantity at each point in space and time around it. A **scalar field** only assigns a **value.** A **vector field** assigns a **value** and a **direction.**

The most common example of a field would be a weather forecast on television. For example, the temperature at various locations across Victoria would be a scalar field as we are being told a specific value for temperature at various locations (i.e. points in space). If the temperature at each location was shown over a 24 hour period, then the value of temperature would also change depending on the time of day (i.e. points in time).

Alternatively, we could consider the wind on a weather forecast. Not only would we be shown the strength of the wind, for example 20m/s, but also the direction, for example from the the South. The speed and direction of the wind would change depending where we are in Victoria and what time of day it is (i.e. points in space and time).

For both of these examples it is possible to create mathematical models, albeit very complex ones, in order to describe the field. Luckily for us we will be looking at much simpler models.

1.1.2 Magnetic Fields

1.1.2.1 Shapes of Magnetic Fields

KEY POINT :
When drawing magnetic field lines, always keep in mind that:

- Lines start from the north pole and end at the south pole (the lines actually continue inside the magnet, forming a closed loop).
- The tangent to a field line at any point gives the direction of the field at that point.
- The direction of a field is the direction of the force acting on a north pole, such as a small compass, at that point.
- The stronger a magnetic field is, the closer the lines are.
- The lines of a magnetic field never intersect.

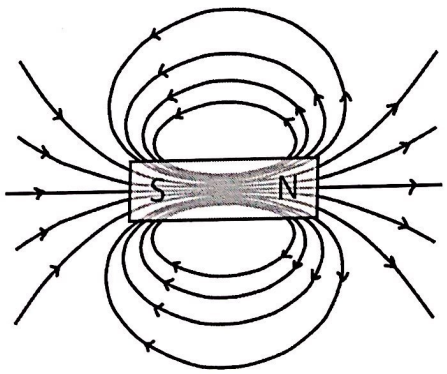

Figure 1.1: Field lines of a bar magnet

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Figure 1.1 on page 2 shows the field lines of a typical bar magnet. The key things to note are that the magnetic field lines never cross; they leave the north pole and enter the south pole; and they are much closer near the ends of the magnet (poles), indicating that the magnetic field is stronger nearer the poles.

The field lines do continue in the magnet, but it is not essential to show these *unless* you are specifically asked to.

1.1.2.2 Right Hand Grip Rule

This is the first of a couple rules that you will need to master before your SACs and exam.

But first, we need to define a new convention for describing fields, as seen in Figure 1.2 on page 3.

KEY POINT :

A dot is used to represent an arrow going into the page.

A cross is used to represent a magnetic field coming out of the page.

The image of an arrow can be used to help you remember: the dot is like the head of the arrow shooting towards you and the cross is like the tail feathers as it shoots away.

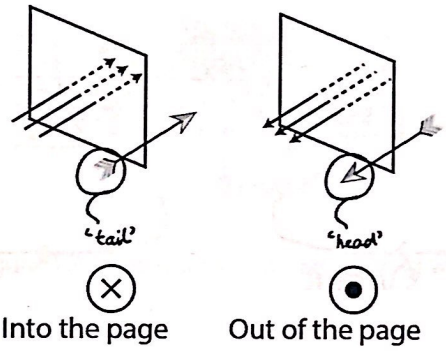

Figure 1.2: Conventions for drawing field lines in 3D

Now to the right hand grip rule, which is sometimes called the *Screw Rule*:

KEY POINT :

For a *straight*, current-carrying wire:

- The thumb points in the direction of the current.
- The fingers curl in the direction of the magnetic field.

The field is stronger closer to the wires, so magnetic field lines are drawn closer.

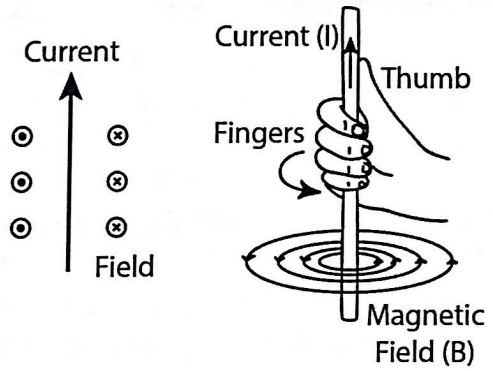

Figure 1.3: Illustration of the right hand grip rule

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1.1.2.3 Right Hand Grip Rule for Solenoids

The right hand grip rule in fact, works in reverse too!

KEY POINT :
For a *looped*, current-carrying wire with many turns,

- The thumb points in the direction of the magnetic field.
- The fingers curl in the direction of the current.

Outside the solenoid, the magnetic field lines are very similar to that of a bar magnet. However, the field lines *do* pass through the solenoid and should be shown.

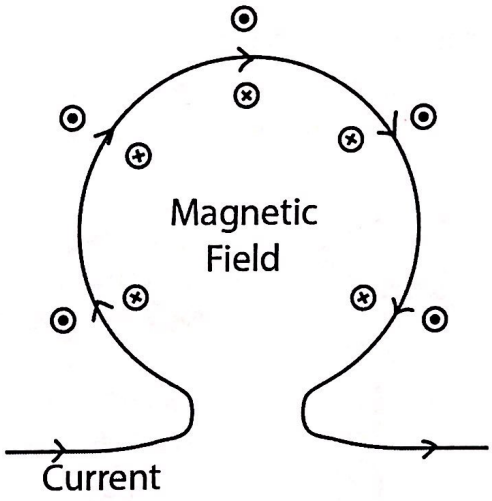

Figure 1.4: Magnetic field lines in a wire loop

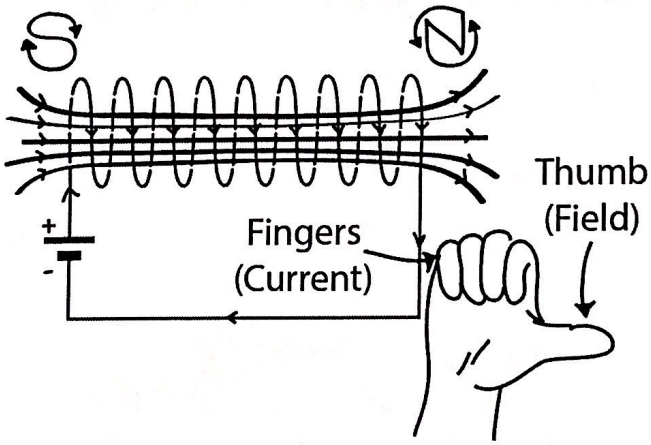

Figure 1.5: Using the right hand grip rule for a solenoid

In Figure 1.4 on page 4 we see a face-on depiction of the coil. Around the outside, the magnetic field is coming out of the page, while the field is going into the page within the coil.

In Figure 1.5 on page 4 we see a side-on depiction of the coil. It looks much like a typical bar magnet, where the north pole is on the right and the south pole is on the left. If you do need to draw the magnetic field lines for a solenoid, you almost always *need* to show the field lines within the coil as well, which is not required for bar magnets.

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1.1.2.4 Right Hand Slap Rule

The right hand slap rule, also known as the *right hand palm rule*, is used to predict the force on a current-carrying wire.

KEY POINT :
For a straight, current-carrying wire:

- The thumb points in the direction of the current.
- The fingers point in the direction of the magnetic field.
- The palm "slaps" in the direction of the force.

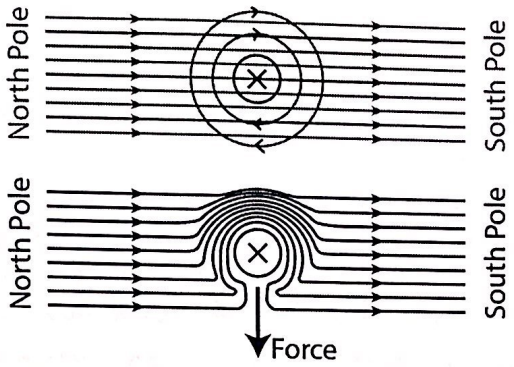

Figure 1.6: Force on a current-carrying wire

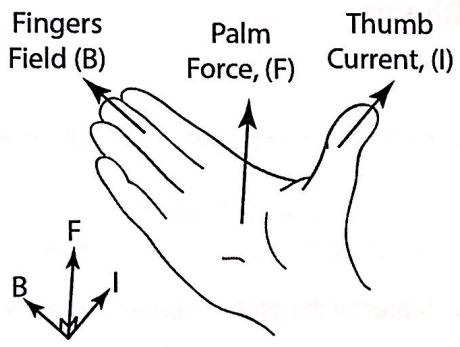

Figure 1.7: Illustration of the right hand slap rule

We know that a current-carrying wire will have its own magnetic field, which we can find using the right hand grip rule. Figure 1.6 on page 5 shows how these fields would look. Originally, on top of the wire (current going into the page) the magnetic field lines act in the same direction so these 'join up'. The lines below the wire are acting in opposite directions, so they would make this unusual 'bunched up' pattern. You will most likely never have to draw this yourself, but it is included so you can see where this force is coming from.

Figure 1.7 on page 5 shows a diagram of the right hand slap rule. Always remember to use your right hand, or you will get the opposite direction for the force if you use your left hand.

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1.1.3 Electric Fields

Key Point:
An electric field is the electric force per unit charge. The direction of the field is is from positive to negative.

For example, suppose there is a positive charge, such as a proton. This proton will have a radially outwards pointing electric field. If there is a nearby proton, it the two will be repelled. If there is a nearby electron, the two will be attracted.

The strength of the electric field is given by:

$$E = \frac{F}{q}$$

...where E is the electric field in N/C or volts/m, F is the electric force in Newtons, and q is the charge in Coulombs.

In order to find the force between two charges, we use Coulomb's Law:

$$F = \frac{kQq}{r^2}$$

...where k is Coulomb's constant 8.99×10^9 Nm^2/C^2, Q is one charge in C, q is the second charge in C, and r is the distance between the centres.

1.2 Effects of Fields

1.2.1 Acceleration on a Charge

Key Point:
The magnitude of the force on a charge, q, moving with a velocity v at an angle θ to the direction of a magnetic field is given by

$$F_B = qvB\sin(\theta)$$

Clearly, the force on the charge is greatest if the path is perpendicular to the direction of the magnetic field.

Key Point:
The magnitude of the force on a charge, q, moving with a velocity v at an angle θ to the direction of a magnetic field is given by

$$F_B = qvB\sin(\theta)$$

1.2.2 Path of Electric Charges

For each electric charge, it is important to keep it in mind that the movement of a charge can be considered as a current.

However, it is especially important to remember that (conventional) current is defined as the movement of a positive charge. Thus, if a positive charge i.e. a proton moves to the right, then this is treated as current flowing to the right. However, if a negative charge i.e. an electron moves to the right, then this is treated as current flowing to the left.

Thus, it is now possible to use the right hand slap rule in order to determine the direction of the force.

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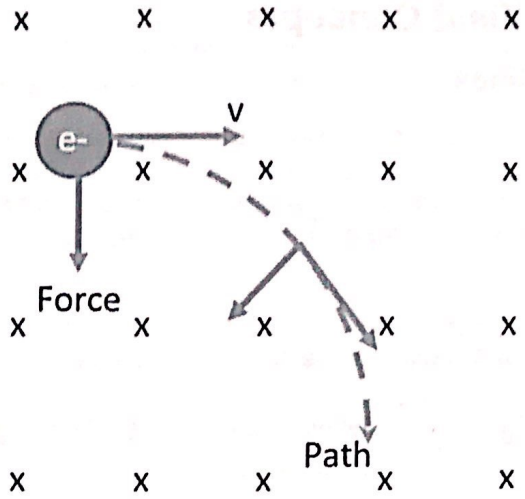

Figure 1.8: Path of a charged particle

Figure 1.8 on page 7 shows the path of an electron in an electric field. Because the electron initially travels towards the right, it is the equivalent of a current moving to the left. Using the right hand grip rule, the force that acts on the charged particle is downwards. This causes the path of the charge to curve in a circle.

Using our knowledge of circular motion, it is possible to relate the force on the charged particle to quantities of the path:

$$F_c = F_{net} = qvB$$

$$\frac{mv^2}{r} = qvB$$

$$\therefore r = \frac{mv}{qB}$$

1.2.3 Gravitational Fields

KEY POINT :
A gravitational field is the force of gravity per unit of mass. The direction of the field is towards the centre of the mass.

For example, consider the Earth and the Sun. There is a gravitational attraction between the two, meaning the Earth is attracted to the Sun and vice versa.

The strength of the gravitational field is given by:

$$g = \frac{F}{m}$$

...where g is the gravitational field strength (m/s^2 or N/kg), F is the force due to gravity (N), and m is the mass of the object (kg).

In order to find the force on two charges, we use Newton's Law of Universal Gravitational Attraction:

$$F = \frac{GMm}{r^2}$$

...where G is the gravitational constant 6.67×10^{-34} m^3/(kg.s^2), M is the mass of the first object (kg), m is the mass of the second object (kg), and r is the distance between the two centres of the two objects.

Notice the difference between these formulae and those of an electric field!

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1.3 Application of Field Concepts

1.3.1 Gravity and Satellites

KEY POINT :
All objects attract each other, but acceleration towards each other *need not* be the same. The attraction (gravity) force always acts *from the centre of mass*, i.e. the core of the Earth, not its surface.

1.3.1.1 Gravitational Fields

For two bodies we can find the gravitational force as:

$$F_g = \frac{GMm}{r^2} \quad where\ G = 6.67 \times 10^{-11} Nm^2 kg^{-2}$$

...where F_g is the force due to gravity, G is the universal gravitational constant, M is the mass of the first body (usually the bigger one), m is the mass of the second body (usually the smaller one), and r is the distance between the *centres* of the two bodies.

When we use $g = 10 m/s^2$ in the equation $W = F_g = mg$ this is a special circumstance when the object is close to the Earth's surface.

EXTENSION :
By way of a quick explanation of why $g \approx 10$ is that it is the result when we plug in the numbers for an object on the surface of the Earth: (note that the Earth isn't exactly spherical, its radius is different in different places, so we use the accepted average radius)

$$F_g = \frac{GMm}{r^2}$$

$$W = \frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times m}{(6.37 \times 10^6)^2}$$

$$mg \approx m \times 9.8$$

From the formula for the Universal Gravitation, it's clear that as the distance between the *centre* of two bodies, r, increases the attraction due to gravity, F_g, decreases as we are dividing by a bigger number. All else equal, doubling the distance r would reduce the force due to gravity F_g by a quarter. This is known as an *inverse square law.*

$$F_g \propto \frac{1}{r^2} \iff \frac{F_{g,2}}{F_{g,1}} = \left(\frac{r_1}{r_2}\right)^2$$

An alternate way to express the strength of gravity is to state the *Gravitational Field Strength.*

KEY POINT :
The gravitational field strength is the force per *unit* mass due to gravity, measured in *N/kg*. To calculate the gravitational field strength, simply divide the force by mass (gravitational field strength is equivalent to saying *the acceleration due to gravity*)

$$g = \frac{F_g}{m} = \frac{GM}{r^2}$$

Note that m/s^2 and N/kg are equivalent units, meaning that they are interchangeable.

As with the force due to gravity, F_g, the gravitational field strength *decreases* as the distance r *increases.* It is also an *inverse square law* relationship:

$$\frac{g_2}{g_1} = d\left(\frac{r_1}{r_2}\right)^2$$

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1.3.1.2 Satellites

A satellite can be any body that orbits a planet, including the moon. At a VCE level, we only consider satellites with no propulsion system in a perfectly circular orbit.

A special type of satellite is known as a **geosynchronous satellite,** sometimes also called a ***geostationary satellite.*** A geosynchronous satellite is unique due to the fact that it remains above the same point on Earth for its entire orbit. This means its period is the same length as a day, $24hrs$ or, since period is in seconds, $86400s$.

KEY POINT :
Since satellites are in free fall, the only force acting on them is the force due to gravity which acts towards the central body. Since satellites travel on a circular path, the *net force* acting them is equal to the centripetal force.

$$\Rightarrow F_c = F_{net} = F_g$$

So, to solve for a satellite's velocity, period or other quantity, the relation $F_c = F_g$ can be used.

This relation $F_c = F_g$ can be further expanded to give what is known as Kepler's Law:

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2} = const.$$

where r is the radius in m, T is the period in s, G is the gravitational constant and the *const.* is constant *only* for satellites orbiting the same central body.

Kepler's Law allows much quicker solving for satellites because much of the formula manipulation has already been done. However, it also tells us that for multiple satellites orbiting *the same body*, the ratio r^3/T^2 is constant. So, if one satellite was orbiting at a radius double the length of the other, its period would be 2.83 times longer.

EXTENSION :
A demonstration of where Kepler's Law comes from (apart from, well, Kepler since it's named after him):

$$F_c = F_g$$

Substituting in our formulas for F_c and F_g

$$\frac{4\pi\pi^2 r}{T^2} = \frac{GM\pi}{r^2}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

The right hand side is constant if all the satellites are orbiting *the same* planet. Otherwise, the value for M would change and the right hand side would no longer be constant.

You'll also notice that interestingly the the *mass of the satellite cancels*, meaning that its path is *not* influenced by its mass.

Note that circular motion (the right hand side) will be covered in much more detail later on!

If we substitute in different formulas for F_g then we can arrive at different equations of motion. Note they are all independent of the mass of the satellite! Some of the more useful ones are listed here:

$$a = g = \frac{GM}{r^2} = \frac{v^2}{r} = \frac{4\pi r}{T^2}$$

$$v = \frac{2\pi r}{T} = \sqrt{\frac{GM}{R}} = \sqrt{rg}$$

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1.3.2 Interactions of Magnets

Hopefully at some time during your life, you've had the joy of playing with magnets, trying to force two north poles together, pulling a north pole and a south pole apart or using magnets to pick up paper clips. We will look a bit further into why these phenomena occur.

KEY POINT :
When drawing field lines or thinking about how magnets will act, it is good to keep in mind:

- Interactions of magnets are based on their magnetic fields, which can be visualised by drawing magnetic field lines.
- Like poles repel and opposite poles attract.
- The Earth is in fact a giant magnet, but it's magnetic poles are the *opposite* of its geological poles. So, the geological North Pole is a magnetic south pole, which is why the North end of a compass points towards it and vice versa for the geological South Pole.

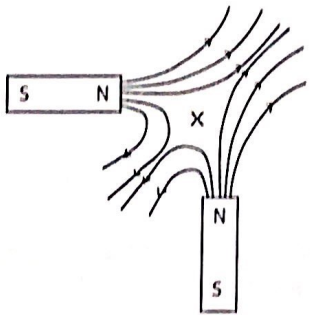

Figure 1.9: Interactions of two magnets

The above image shows two north poles acting at an angle to each other. Since the field lines *do not* cross, when they come close to each other, they 'bend away' from each other. So, at around point X, the magnetic field will actually be zero! This point has been rather exaggerated, but it will occur in real life even if it is a tiny point. The general idea is to get you thinking about how field lines will behave.

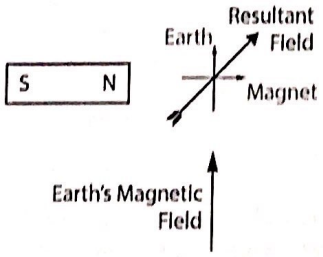

Figure 1.10: Vector sum of field lines

This image shows the Earth's magnetic field and a bar magnet. Now, we could draw field lines, but another way to find the magnetic field is to use vector addition. At the marked point, the magnet would have a magnetic field acting to the East, and the Earth's field would be acting to the North. So, when we add these two together the resultant field is pointing North East, as shown.

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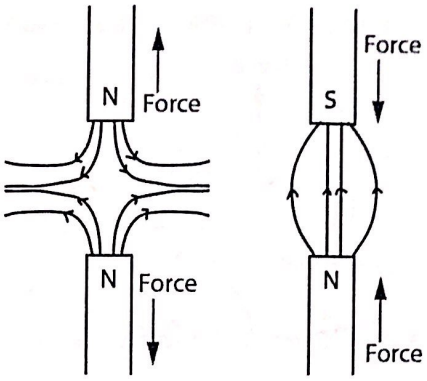

Figure 1.11: Magnetic fields of like and opposite poles

This figure shows two like poles and two opposite poles. For the like poles, it is clear that the magnetic field lines have to 'bend away' from each other, as magnetic field lines *never* cross other lines. Because of this interaction of field lines, we have the outwards pushing force. For the opposite poles, the field lines will actually 'join up' as they can form continuous lines, instead of having to 'bend away' from each other. So, they will have an inwards pushing force.

This is just a general idea of why likes repel and opposites attract! You don't need this for your exam, but you may be called upon to draw the magnetic field lines or describe the direction of the magnetic field at a specified point.

1.3.3 DC Motors

1.3.3.1 Force on a Current-Carrying Wire

We know the direction of the induced force, so we also need a method of calculating the magnitude of the induced force.

KEY POINT :
The magnitude of a force on a current-carrying wire induced by a magnetic field is given by:

$$F = nBIL$$

...where F is the resultant force (measured in newtons, N), n is the number of coils of wire, B is the magnetic flux density (measured in tesla, T, where T = N m^{-1} A^{-1}), I is current (measured in amps, A), and L is the length of the wire (measured in metres, m).

1.3.3.2 Rotation of the Coil (Armature)

A direct current (DC) motor is essentially a coil of wire in a near-constant magnetic field. There are more complicated motors available, many of which have multiple 'arms' for efficiency, but we only need to look at a single flat coil for VCE.

KEY POINT :
When the coil is initially in the horizontal position, the forces on the each side side of the armature act in *opposite* directions due to the magnetic field. This causes a turning effect (torque).

To determine the direction of rotation, we use the right hand slap rule for each side.

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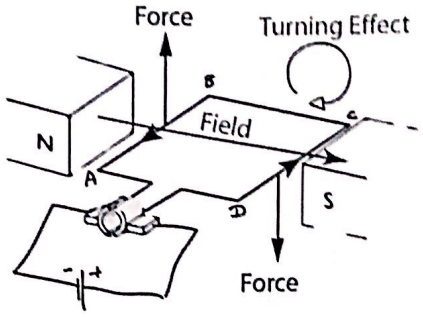

Figure 1.12: Turning effect on a coil in a DC motor

Example 1.1

Consider the coil ABCD in Figure 1.12 on page 12. Suppose that each of the sides AB, BC and CD have a length of 0.8 m. If the magnetic field has a strength of 0.5 T and the current flowing is 1.5 A, calculate the force on the sides AB, BC and CD.

Side AB:

$$F = BIL \rightarrow F = 0.5 \times 1.5 \times 0.8 = 0.6\,\text{N}$$

Side BC:

The current is parallel to the field, so the force is zero.

Side CD:

$$F = BIL \rightarrow F = 0.5 \times 1.5 \times 0.8 = 0.6\,\text{N}$$

So, the forces are: 0.6 N upwards on the side AB, 0 on the side BC and 0.6 N downwards on the side DC.

1.3.3.3 Function of a Commutator (Split Ring)

KEY POINT :

As the coil rotates between the horizontal and vertical position as shown in Figure 1.13 on page 12:

- The magnitude of the force on the coils stays then same, since $F = nBIL$ and n, B, I, L remain constant.
- The direction of the forces on the sides of the coils stay the same i.e. the force on the side rotating up still acts upwards and the force on the side rotating down still acts downwards.
- The turning effect (torque) decreases as the coil rotates and is zero when the coil is vertical, since the side on the top wants to go up and the side on the bottom wants to go down. At this point, the coil would oscillate and come to a stop in the vertical position.

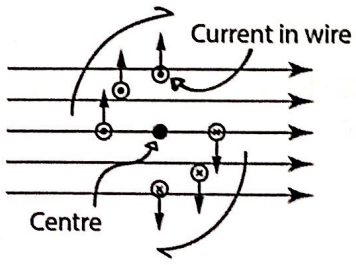

Figure 1.13: Rotation of a coil in a field

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So how do we make sure that the coil keeps on turning? We use a commutator to flip the direction of the current when needed.

KEY POINT :
The role of a split-ring commutator is to reverse the direction of the current in the coil to keep the motor turning in the same direction. It achieves this because:

- The coil is connected to the power supplies by the use of brushes, so the connections from the coil to the power supply can be 'flipped' without any tangling of wires as shown in Figure 1.14 on page 13.
- As the coil swings through the vertical point, the connections are reversed so the current flows in the opposite direction.
- The reversal of current means that, according to the right hand slap rule, the direction of the forces are reversed, i.e. the side on top will want to continue rotating downwards and vice versa for the side on the bottom.

This reversal of current has to occur every half turn when the coil passes through the vertical position.

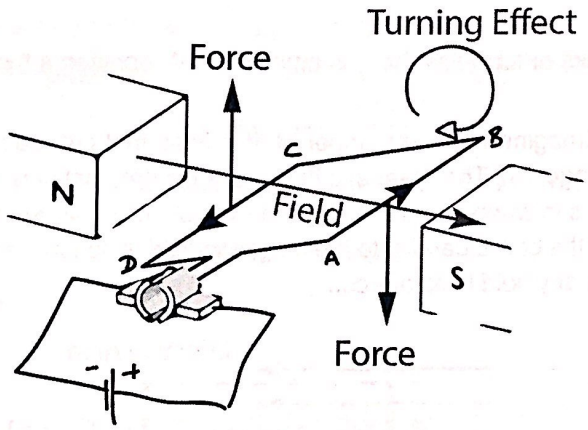

Figure 1.14: Purpose of a (split ring) commutator

Almost *every* year there is a question in the exam about the use of a commutator in a DC motor. A definition like the one above doesn't take up that much space on your cheat sheet, especially if you type it up, and can provide you with a safety net for your exam.

1.3.4 Particle Accelerator

A particle accelerator exerts a force on a particle to speed it through a field.

KEY POINT :
Electrons can be accelerated by an electric field between two parallel charged plates (one positive, one negative). The force is given as:

$$F = qE = \frac{qV}{d}$$

...where q is the charge on the particle, E is the field strength, V is the potential difference between the plates, and d is the distance between the two plates.

In addition, the change in kinetic energy if the particle starts at rest is given by:

$$\Delta KE = qV = \frac{1}{2}mv^2$$

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Area of Study 2

How are fields used to move electrical energy

2.1 Generation of Electricity

2.1.1 Magnetic Flux

Whilst B measures the magnetic field intensity, there is another quantity that interests us: **magnetic flux.**

KEY POINT :
Magnetic flux is defined as:

$$\Phi = B A_{\perp}$$

...where Φ is the magnetic flux (measured in T m^2 or Weber, Wb), B is the strength of the magnetic field (measured in tesla, T) and $A_{\perp}$ is the *effective* area perpendicular to the field (measured in square metres, m^2).

Note that in many textbooks or formulas the $\perp$ is omitted, so A represents the effective area.

The effective area can be imagined as the number of field lines that cut (pass through) the coil, and can be seen in Figure 2.1 on page 14. The magnetic flux is *always* greatest when the coil is perpendicular to the magnetic field, since this is when $A_{\perp}$ is a maximum - the effective area is just the area of the coil. The magnetic flux is *zero* when the coil is parallel to the magnetic field, since $A_{\perp} \approxeq 0$ as only the side of the coil points into the field - barely any field lines are cut.

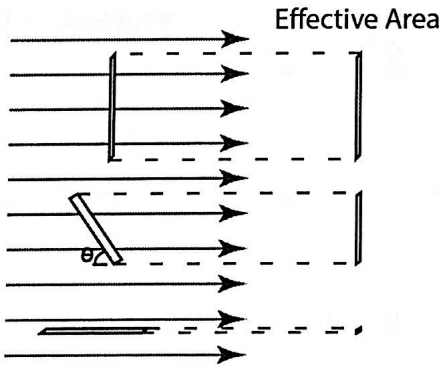

Figure 2.1: Effective area

EXTENSION :
The magnetic flux can be found by calculating:

$$\Phi = B A_{\perp} = BA\cos(\theta)$$

...where Φ is the magnetic flux (measured in T m^2 or Weber, Wb), B is the strength of the magnetic field (measured in tesla, T), A is the area of the coil (measured in square metres, m^2) and θ is the angle between the field lines and the coil. This is interesting because it shows us how to calculate the magnetic flux of a rotated coil, but is *not* required by the study design.

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2.1.2 Induction

2.1.2.1 Faraday's Law

This is a key law that can be expressed both in words or maths symbols, so you should try to familiarise yourself with both!

KEY POINT :
Faraday's Law states that: *"The magnitude of the induced electromotive force (EMF) is equal to the negative rate of change of the magnetic flux."*

The average EMF is given by:

$$\varepsilon_{average} = \frac{-\Delta\Phi}{\Delta t} = -\frac{\Delta(BA)}{\Delta t}$$

...where $\varepsilon_{average}$ is the average EMF induced (measured in volts, V), $\Delta\Phi$ is the *change* in magnetic flux (measured in T m^2), Δt is the *change* in time (measured in seconds, s).

In the second equation, we substitute in $\Phi = B\,A$ to show that a change in Φ can be caused in a change in magnetic field strength B and/or a change in the effective area A.

For a rotating coil, there are several interesting features:

- The formula for n coils is given as:

$$\varepsilon_{average} = \frac{-n\Delta\Phi}{\Delta t}$$

- The minus sign (–) indicates the direction that the current will flow in (recall that conventional current flows from positive to negative).
- The shorter the time taken, the larger the EMF produced.

2.1.2.2 Flux-Time Graphs

KEY POINT :
Given a magnetic flux vs time graph, we can apply Faraday's law:

$$\varepsilon_{average} = \frac{-\Delta\Phi}{\Delta t} = \frac{-\text{rise}}{\text{run}} = \text{negative gradient of flux vs time graph}$$

So, we can see that only when EMF is changing, i.e. the gradient is not zero, there is an EMF produced.

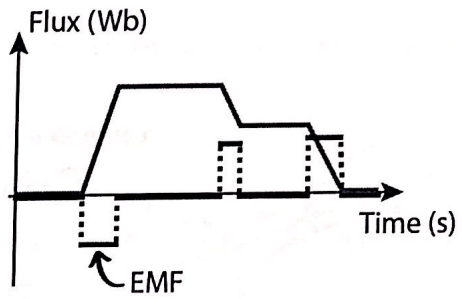

Figure 2.2: Example of a flux vs time graph

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Example 2.1

For Figure 2.2 on page 15, explain the relation of the flux and EMF produced. Refer to any relevant laws.

The relevant law we will use is Faraday's Law: $\varepsilon_{average} = \frac{-\Delta\Phi}{\Delta t}$. We can see that the EMF produced is essentially the negative gradient of the flux vs time graph:

- When the flux is constant, the change in flux is zero. So the EMF is zero.
- When the flux is increasing, it has a positive gradient. So, the EMF is negative.
- When the flux is decreasing, it has a negative gradient. So the EMF is positive.

The first EMF spike is the largest, because the gradient (rate of change) of the flux is the largest. The second spike is the smaller because the gradient (rate of change) is smaller.

2.1.2.3 Lenz's Law

Lenz's law appears relatively simple on the face of things, but you need to take care when solving these questions as they can become quite complicated.

KEY POINT :

Lenz's Law states that: *"The induced current in a loop will be in such a direction that its magnetic flux will oppose the change in magnetic flux that produced it".*

It is a specific use of conservation of energy and Newton's 3rd Law. We use Lenz's Law in combination with the right hand grip rule in order to find the direction of the current in a loop.

2.1.2.4 Solving Using Flux

KEY POINT :

The following table outlines the 'opposing flux' to some initial change in flux. The opposing flux could be *either* reversing the increase/decrease *or* acting in a different direction.

Initial change in flux	Opposing Flux
Increasing to the right	Decreasing to the right or **Increasing** to the left
Increasing to the left	Decreasing to the left or **Increasing** to the right
Decreasing to the right	**Increasing** to the right or Decreasing to the left
Decreasing to the left	**Increasing** to the left or Decreasing to the right

Given some change in flux through a coil, we always want to use the option where flux is *increasing*, because the current in the loop cannot induce negative or decreasing flux. These are the ones in bold in the table above.

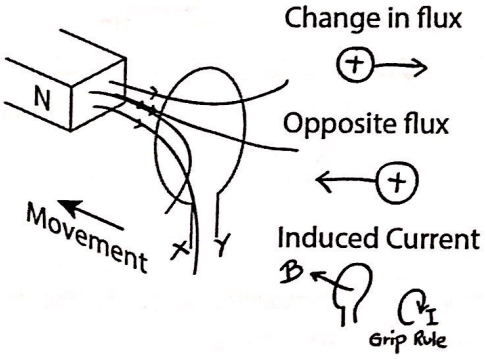

Figure 2.3: Using flux to solve Lenz's Law

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Figure 2.3 on page 16 shows the general procedure for solving this question. The process is:

1. Identify the change in magnetic flux:
   - Direction of magnetic flux: Right (→)
   - Increase or decrease: Increase (+)
2. Identify the opposing flux:
   - Change the direction: Increasing to the left
3. Identify the direction of the current:
   - Right hand grip rule: Clockwise current
4. State your answer:
   - The induced current travels clockwise from X to Y

2.1.2.5 Solving Using Magnetic Poles

KEY POINT :
When we have a magnet that is moving relative to the coil, we can find the pole that is induced in the coil facing the magnet. This pole must oppose the motion of the magnet (Lenz's Law).

Movement of magnet	Magnetic Pole (induced) facing the magnet		
N pole closer to loop	Oppose: closer	Like poles generated	N pole
S pole closer to loop	Oppose: closer	Like poles generated	S pole
N pole further from loop	Oppose: further	Opposite poles generated	S pole
S pole further from loop	Oppose: further	Opposite poles generated	N pole

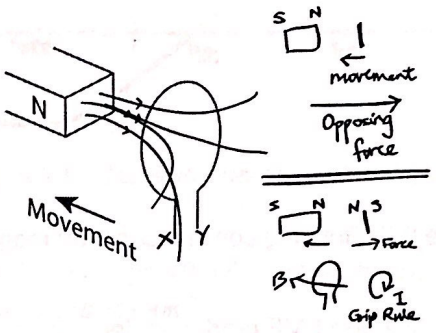

Figure 2.4: Using poles to solve a Lenz's Law question

Figure 2.4 on page 17 shows the general procedure for solving this question. The process is:

1. Identify movement
   - Closer to or further away: Closer to
   - North or south pole: North
2. Identify opposing force
   - Magnetic force opposes the force pushing the loop towards the magnet
   - Attraction or repulsion: Repulsion
3. Identify the pole on the wire facing the magnet
   - Like or opposite poles: For repulsion, like poles
   - North or south pole: North
4. Identify the direction of the current
   - Right hand grip rule: Clockwise current
5. State your answer
   - The induced current travels clockwise from X to Y

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2.1.3 Generators

The general fundamental idea of electricity generation is Faraday's law, as it can predict the EMF produced. Recall that Faraday's law is essentially the rate of change of flux, $\frac{\Delta\Phi}{\Delta t}$, and to change flux we need to change the effective area or the magnetic field strength. It is *much* easier to change the effective area, which is what is done in modern generators.

Key Point :
A consistent method of generating electricity, by varying flux, is to rotate a coil in a magnetic field.

- The flux in the coil will vary periodically.
- The flux vs time can be plotted and Faraday's law can be used to find the EMF vs time graph.
- The signal produced is a sinusoidal wave.

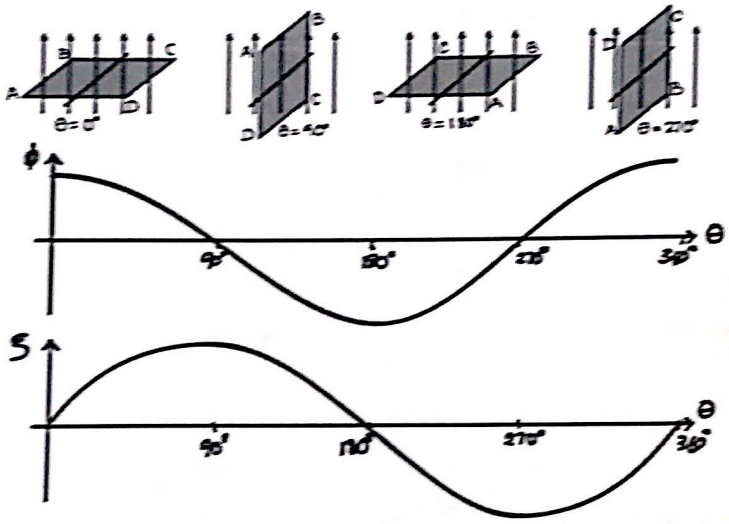

Figure 2.5: Rotating coil in a constant magnetic field

Figure 2.5 on page 18 shows the flux and EMF produced depending on the rotation of the coil. Initially, the coil ABCD is perpendicular to the field, so it has a *maximum positive* magnetic flux. However, as it rotates, the effective area becomes smaller and smaller, until it is zero. At this point, the magnetic flux will be zero since $\Phi = BA$. As it continues to rotate, the magnetic field is passing through the 'other side,' DCBA so the EMF produced is going to be negative. When the coil has rotated through 180 degrees the flux is now a *maximum negative* magnetic flux. The coil continues to rotate until it is again parallel to the field and has an effective area of zero. As it continues to rotate, it has a 'positive side' again, ABCD, so the magnetic flux is positive again. And so the cycle continues.

Note that as the flux is changing, we could in fact use Lenz's Law to determine the current, if it flowed A→B→C→D or the other way around. But in fact the EMF graph tells us how the current will behave.

The EMF graph is found by using Faraday's Law, $\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$; imagine finding the gradient at various points on the flux vs. time graph to plot the EMF graph. We could have differentiated the function for flux, but it is *not* required for VCE Physics. We see that for half a turn, the EMF is positive and the other half it is negative. This means that the positive and negative must have switched... meaning the current in the coil has changed direction as well! So, when the coil turns in the magnetic field, for half the turn the induced current flows one way, and then the other way for the next half turn. The direction of the current changes when the coil is ***perpendicular*** to the field.

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2.1.3.1 Function of a Alternator (slip rings)

KEY POINT :
The coil needs to rotate freely without a mess of tangled wires.

- The induced current in the coil is transferred to the external circuit through carbon brushes rubbing against slip rings.
- The output is AC (alternating current) because the voltage (and current) act in one direction for half a cycle then act in the other direction for the other half of the cycle.
- This 'alternating' can be seen in a EMF vs time graph.

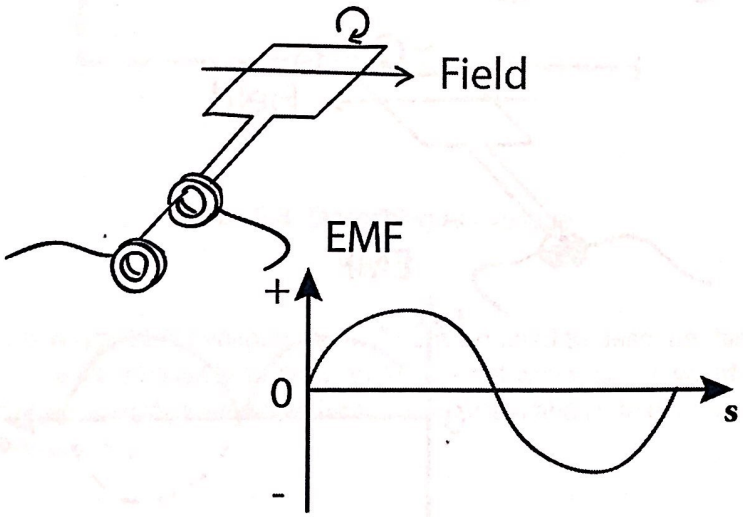

Figure 2.6: Generator using an alternator

Example 2.2
For the generator pictured in Figure 2.6 on page 19, what would happen if the coil was rotated at double the speed? How would the EMF vs time graph look?

The first thing that would happen is that at double the speed (i.e. double the frequency), the EMF graph would have half the original period. So, it would cycle from peak to trough at double the speed; taking half as long to complete a full cycle. The second thing that would happen is that the induced EMF would double, according to Faraday's law. This is because $\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$ and Δt has been halved.

So the graph of EMF vs. time would be twice as tall but half as wide.

EXTENSION :
The peak output voltage can be given by:

$$\varepsilon_p = nBA\,2\pi f$$

...where ε_p is the peak voltage (measured in volts, V), n is the number of coils, B is the strength of the magnetic field (measured in tesla, T), A is the area of the coil (measured in square metres, m^2) and f is the frequency of rotation (measured in hertz, Hz). However this is *not* required in the current study design.

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2.1.3.2 Commutators in Generators

KEY POINT :
No matter the output connections, the current in the coil is always AC.

However, if a commutator is used, as the current in the coil reverses (every half turn) the output terminal connects are reversed too (every half turn).

- The output is pulsating, unidirectional (DC) voltage.
- The peak voltage is the same as when an alternator is used.
- Essentially the same as a DC motor, but in reverse.

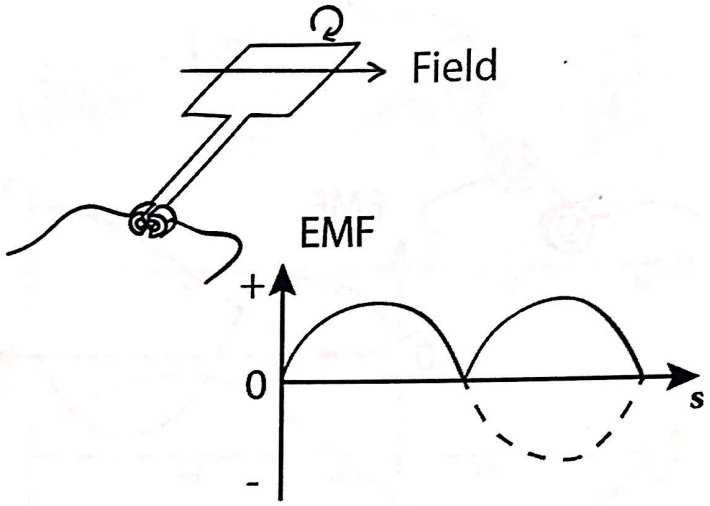

Figure 2.7: Generator using a commutator

As you may have noticed, all generators operate under the principle of a rotating coil. Hydroelectric dams use water to turn turbines to drive generators. Coal and gas are burnt to heat water to steam which is used to turn turbines which drive generators. Uranium undergoes fission releasing heat that is used to heat water to steam to turn turbines. There is nothing 'special' about coal or uranium, they are just very good at heating water!

2.2 Transmission of Electricity

2.2.1 AC Voltage

We begin by defining some key terms and measurements for a typical AC wave, such as that generated in a generator using an alternator.

- **Period (T):** is the time taken for a complete cycle.
  - Example: in Australia, this is 0.02s.
- **Frequency (f)** is the number of cycles in one second.
  - Example: in Australia, this is 50Hz.
- **Relation of period and frequency**: these two quantities are related by: $f = \frac{1}{T}$.
- **Peak voltage (V_p):** the amplitude of the wave, i.e. the highest point.
  - Example: in Australia, this is 339 V.
- **Peak-to-peak voltage (V_{p-p}):** the vertical distance between the highest and lowest point. V_{p-p} can also be found by $V_{p-p} = 2V_p$.

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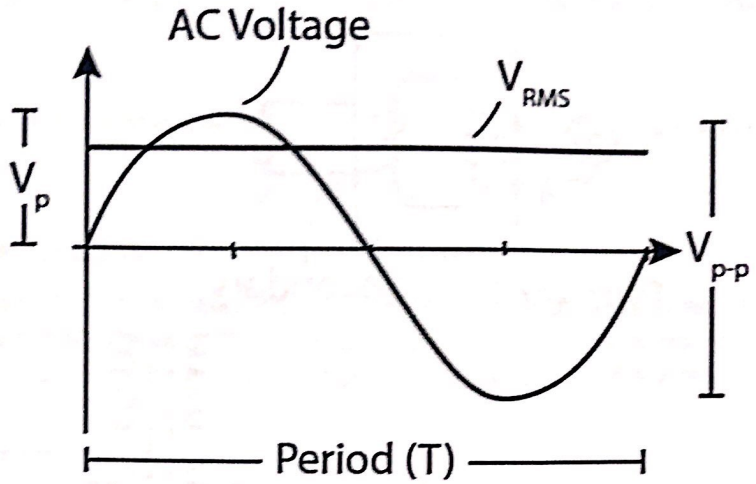

Figure 2.8: Describing AC Voltage

Figure 2.8 on page 21 is a typical AC voltage wave. The word used to describe this shape is 'sinusoidal' (think of a sine graph). It is also possible to draw an AC current wave, and it would have much the same properties, but instead of peak voltage we would have peak current and instead of peak-to-peak voltage we would have peak-to-peak current.

KEY POINT :
A much more useful description of AC voltage is the root mean square (RMS) value. This is the *equivalent* DC voltage that would have the same average power as the AC voltage.

Example: In Australia the peak voltage is 339V and the RMS value is 240V.

RMS voltage can be thought of as the average voltage.

$$V_{RMS} = \frac{V_p}{\sqrt{2}} = \frac{V_{p-p}}{2\sqrt{2}}$$

$$I_{RMS} = \frac{I_p}{\sqrt{2}} = \frac{I_{p-p}}{2\sqrt{2}}$$

$$P = \frac{V_{RMS}^2}{R} = V_{RMS} I_{RMS}$$

2.2.2 Transformers

KEY POINT :
A transformer works on the principle of changing magnetic flux.

- A changing current in the primary coil causes a changing magnetic field.
- There is a changing magnetic flux in the core.
- A voltage is induced in the secondary coil because of this changing flux in the core.

It is generally assumed that energy is conserved, i.e. the input power is equal to the output power.

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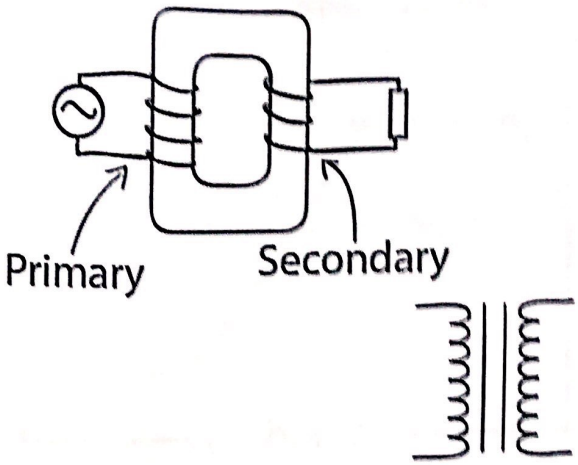

Figure 2.9: Transformer and its symbol

Example 2.3

For the transformer shown in Figure 2.9 on page 22, two students are debating what will happen when a DC battery is used instead of an AC power supply. One student predicts nothing will happen to the voltage in the secondary coil after the battery is connected, while this other says there will be spike of voltage in the secondary coil. Which student is correct?

Both students are correct! When the battery is connected, the current in the loop changes from zero to a higher value. Since the current has changed, the magnetic field has changed which causes a change in magnetic flux. So, at the instant the battery is connected, there is a changing flux in the coil which, according Faraday's Law, will induce a voltage in the secondary coil. So the second student is correct about the spike in voltage.

Now, when the battery is connected, there will no longer be any change in voltage or current. So, the magnetic field is constant, meaning there is no longer a change in flux. As the change in flux is zero, this means that there will be no induced EMF in the secondary current (Faraday's law). So, the first student is correct about the zero voltage *after* the battery is connected.

KEY POINT :

The voltages in the primary and secondary coils can be related to the number of turns of wire:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

For an ideal transformer, we assume that there is no loss in energy, so:

$$P_p = P_s \longrightarrow V_p I_p = V_s I_s$$

This means that we can also find the ratio of currents:

$$\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

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There are two types of transformers:

- A step-up transformer is where the output voltage is greater than the input voltage (the output current is smaller):

$$N_s > N_p \qquad V_s > V_p, I_s < I_p$$

- A step-down transformer is where the output voltage is less than the input voltage (the output current is larger):

$$N_s < N_p \qquad V_S < V_p, I_s > I_s$$

2.2.3 Transmission Loss

Now how do we physically transmit electricity? Wires of course! But unless we are using some special supercooled ultra low resistance wires that probably cost more than the power loss they save, there will always be some power lost in the transmission lines due to resistance in the wires.

KEY POINT :

The current in the line is given by:

$$I_{wires} = \frac{P_{in}}{V_{in}}$$

The voltage drop in the lines is given by:

$$V_{drop} = I_{wires} R_{wires}$$

So, the output voltage is given by:

$$V_{out} = V_{in} - V_{drop}$$

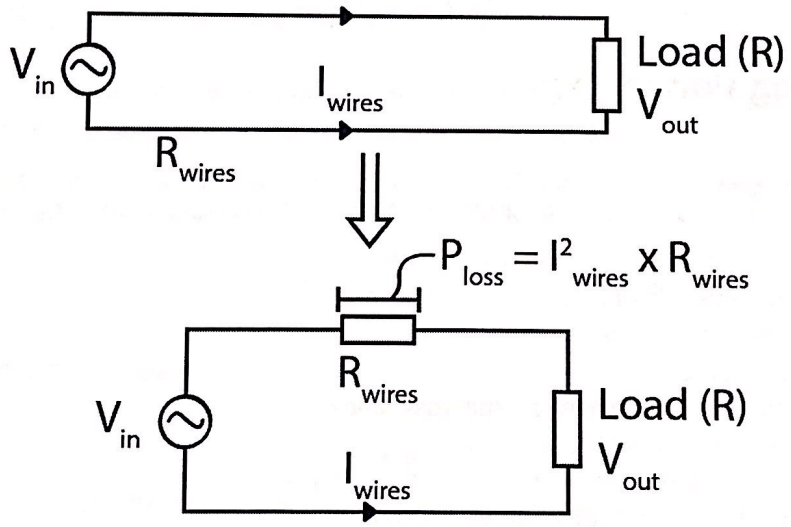

Figure 2.10: Basic voltage drop and power loss

Figure 2.10 on page 23 shows how we can 'condense' the resistance of the wires into an actual resistor to simplify our calculations. So we assume the resistor has a resistance equal to the total resistance of the wires in the original circuit.

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Example 2.4

A farmer has two wires, each with an internal resistance of 5.0 Ω. He uses these wires to hook his 800 W generator to his house. If the generator provides a voltage of 250 V, find the voltage at the farmer's house.

Our first step is to find the *total* resistance of the wires,

$$R_{wires} = 2 \times 5 = 10\ \Omega$$ (There are *two* wires.)

Now we can find the current in the wires,

$$I_{wires} = \frac{P_{in}}{V_{in}} \rightarrow I_{wires} = \frac{800}{250} = 3.2\,\text{A}$$

Find the voltage drop in the wires,

$$V_{drop} = I_{wires} \times R_{wires} = 3.2 \times 10 = 32\,\text{V}$$

We can now find the voltage at his house,

$$V_{out} = V_{in} - V_{drop} = 250 - 32 = 218\,\text{V}$$

We could also continue on to find the power loss in the wires.

KEY POINT :

The current in the line is given by:

$$I_{wires} = \frac{P_{in}}{V_{in}}$$

The power loss in the lines is given by:

$$P_{loss} = I_{wires}^2 R_{wires}$$

So, the output power is given by:

$$P_{out} = P_{in} - P_{loss}$$

2.3 Reducing Power Loss

KEY POINT :

To reduce power loss, either the current or the resistance in the wires can be decreased.

- Reduce resistance: less effective as $P_{loss} \propto R_{wires}$
  - Use wires with a thicker cross-sectional area
  - Use a shorter distance
  - Use more wires
  - Use materials with smaller internal resistance
- Reduce current: more effective as $P_{loss} \propto I_{wires}^2$
  - Reduce input power (same voltage but less current)
  - Same power (more voltage but less current)

The best method of reducing the current is to use transformers. Since transformers rely on AC voltage, this is primarily why DC voltage is rarely used to transfer electricity.

KEY POINT :

A pair of step-up and step-down transformers can be used to effectively reduce power loss:

- Power is generated at about 20kV and then it is stepped up to 400kV via a transformer.
- The current in the transmission lines is $\frac{1}{20}$ of the previous value.
- The power loss is $\left(\frac{1}{20}\right)^2 = \frac{1}{400}$ of a system with no transformers.
- Near the consumer a step-down transformer is used to step down the voltage to 240V.

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Area of Study 3

How fast can things go?

3.1 Laws of Motion

3.1.1 Newton's First Law

KEY POINT :
Definition: *"Every object continues to be at rest, or continues with constant velocity, unless it experiences an unbalanced force."*

In plain English, Newton's First Law is an explanation of why objects can travel through space even though *no force* acts upon them. For example, when a tennis ball is thrown (ignoring air resistance) it will continue on even though it has left the thrower's hand. This is the first half of the definition.

When an unbalanced force acts upon this tennis ball, namely gravity, the motion of the ball is altered so it travels in an arc instead of a straight line. An unbalanced force is the equivalent of a *non-zero net force* (modern definition). This is the second half of Newton's First Law.

This law is also known as *The Law of Inertia*.

3.1.2 Newton's Second Law

KEY POINT :
Definition: *"The acceleration of a body experiencing an unbalanced force is directly proportional to the net force and inversely proportional to the mass of the body."*

This is a very wordy summary of Newton's experiments. He found that the acceleration increased *linearly* as the net force increased and decreased *linearly* as the mass increased.

A mathematical representation of this phenomenon is: $a \propto \Sigma F$ and $a \propto \frac{1}{m}$. A much more simple representation is: *force equals mass times acceleration.*

$$\text{Net Force } (\Sigma F \text{ or } F_{net}) = \text{Mass } (m) \times \text{Acceleration } (a)$$

3.1.3 Newton's Third Law

KEY POINT :
Definition: *"When one body exerts a force on another body (action force), the second body exerts an equal force in the opposite direction on the first (the reaction force)."*

This is one of the more straightforward laws in terms of wording. The important points to keep in mind are that the forces are *equal* but in *opposite* directions acting on *different* objects. Once those are set in your mind, you can simply remember Newton's third law as: ***for every action there is an equal and opposite reaction.***

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3.1.4 Circular Motion

3.1.4.1 Centripetal Acceleration

For any object that is travelling in a circle, the velocity is always tangential to the circle – it wants to 'escape' the circle. As the object travels through the circle clearly the direction of velocity is changing. This indicates that there must be some sort of acceleration present that is changing the velocity *even though the magnitude of velocity (the speed) is constant.* This acceleration is called 'centripetal acceleration' which literally means centre-seeking because the acceleration always points towards the centre.

We can calculate the average speed by using speed equals distance divided by time. Distance is given by the circumference of the circle and the time is the length of time it takes to complete one revolution, known as the period.

$$v = \frac{circumference}{time} = \frac{2\pi r}{T}$$

...where v is in metres per second, r is in metres, and T (period) is in seconds.

The centripetal acceleration is given by:

$$a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$$

...where a is in metres per second^2, v is in metres per second, r is in metres, and T (period) is in seconds.

Another way to express T is the number of complete circles per second, known as the frequency. The relationship is given as: $T = \frac{1}{f}$ where f is in Hz.

3.1.4.2 Centripetal Force

KEY POINT :
The centripetal force is *not* an actual force. It is the *net force*, the sum of all the forces that act on a body that causes it to travel in a circle.

For example, in a roller coaster ride there may be several forces such as track pushing against the bottom of the cart (the normal force), frictional forces and gravitational forces (weight forces). But, when we sum all of these forces together, the *net force* points towards the centre which is the *centripetal* force because it acts towards the centre of circular motion.

The centripetal force should not be confused with the centrifugal force. The centripetal force acts towards the centre causing the circular motion (centripetal is a combination of two Latin words that literally means 'to seek the centre'). On the other hand, the centrifugal force literally means 'to flee from the centre' and is due to the *apparent* force that pushes a body away from the centre because of its inertia. Though this is an important distinction, centrifugal force rarely appears at the VCE level.

Now, since the centripetal force is a net force, Newton's Second Law can be applied to model circular motion. Recall that $F_{net} = m \times a$.

$$\therefore F_{net} = F_c = m \times a_c$$

where F_c is the centripetal force, m is the mass and a_c is the centripetal acceleration

Since we know what the centripetal acceleration is, we can substitute it in for a_c

$$\Rightarrow F_{net} = F_c = \frac{mv^2}{r} = \frac{4m\pi^2 r}{T^2}$$

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3.1.4.3 Solving Horizontal Circles

The key to solving horizontal circular motion is to keep in mind that the centripetal force always acts towards the centre and that it is equal to the net force, $F_{net} = F_c$.

Example 3.1

A race car of mass 1500*kg speeds around a gravel corner at* 30*m/s. If the corner is perfectly flat and can be modelled as a semi-circle of radius* 100*m, find the centripetal force acting on the car and state the source of the centripetal force.*

We can directly use our equations for centripetal acceleration

$$F_c = \frac{mv^2}{r} = \frac{1500 \times 30^2}{100}$$
$$= 13.5kN$$

We know that the centripetal force is the net force and not the actual force. The only force that acts towards the centre of the circle. So, the source of the centripetal force is friction.

KEY POINT :

For banked curve questions, where an object travels in a horizontal circle on a banked track, there is a shortcut formula available if friction is neglected: $\tan(\theta) = \frac{v^2}{gr}$.

To see the 'source' of the centripetal force, consider the image below. It can be seen that the only force that acts towards the centre is the horizontal component of the normal force.

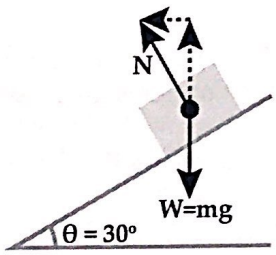

Figure 3.1: Banked curve example

EXTENSION :

To derive the shortcut formula for a banked curve, let up be positive in the vertical direction and left be positive in the horizontal direction:

There is no vertical motion, so in the vertical direction

$$F_{net} = 0 = N\cos(\theta) - W$$
$$\Leftrightarrow N\cos(\theta) = mg \ldots (1)$$

In the horizontal direction, the net force is equal to the centripetal force

$$F_{net} = F_c = N\sin(\theta)$$
$$\Leftrightarrow N\sin(\theta) = \frac{mv^2}{r} \ldots (2)$$

If we divide the two equations, (2) ÷ (1)

$$\Leftrightarrow \frac{\cancel{N}\sin(\theta)}{\cancel{N}\cos(\theta)} = \frac{\frac{\cancel{m}v^2}{r}}{\cancel{m}g}$$
$$\therefore \tan(\theta) = \frac{v^2}{gr}$$

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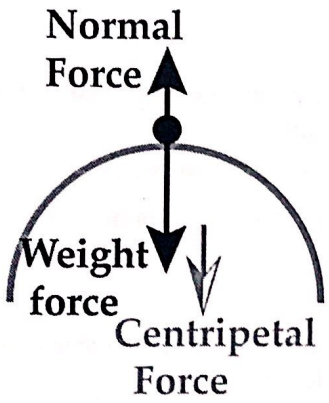
3.1.4.4 Solving Vertical Circles (Highest Point)

Vertical circles are much more complex to solve than their horizontal counterparts because one has to account for the effects of gravity at various points on the circle. Thus, it is only specified that students need to understand the motion at the highest and lowest points of a vertical circle.

First considered is a scenario when an object is at the top of the vertical circle. However, there are two such possible cases. For a roller coaster, the cart could be travelling over the top of a hill, *on top of the track.* Or, the cart could be performing a loop-the-loop and be dangling *on the bottom of the track.* Each case will be considered separately.

Case 1: Outside the Circle



From the diagram, it's evident that there are only *two* forces that act on the object: the normal force and the weight force. Remember that the centripetal force is *not* a force that acts on the object, it is the *net force.*

KEY POINT :
For an object on the 'outside' of the circle on the top, we can write the equation of motion as follows. In this equation, *down* is taken as positive (remember to be consistent with your sign convention!)

$$F_{net} = F_c = W - N$$

$$\frac{mv^2}{r} = mg - N$$

$$\Leftrightarrow N_{(up)} = mg - \frac{mv^2}{r}$$

An interesting scenario, often used in questions, is when the object is travelling *fast* enough to lose contact with the circular path it's travelling on and fly off into mid air. To solve such scenarios, we must understand that the *normal force would be zero* as it leaves the surface because there is no more contact. Therefore, we would plug $N = 0$ into our equation and solve.

EXTENSION :
We can easily derive a shortcut formula for such a scenario, although (as always) I stress the necessity of knowing how to tackle questions from scratch without the need for shortcut formulas.

We know that from above:

$$N_{(up)} = mg - \frac{mv^2}{r}$$

Now plug in $N = 0$,

$$0 = mg - \frac{mv^2}{r}$$

$$\therefore v = \sqrt{gr}$$

Thus, to find the minimum speed for an object to leave the surface we don't actually need to know its mass! All that is required is the radius of the track.

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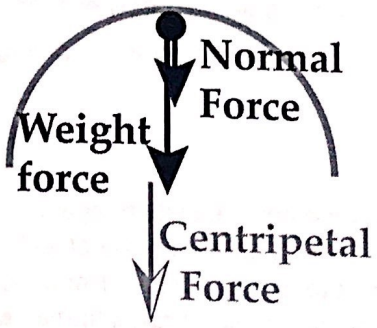
Case 2: Inside the Circle



From the diagram, it's evident that again there are only *two* forces that act on the object: the normal force and the weight force. The normal force is acting downwards because the track is above object, so the contact surface is on top of the object. As the normal force acts perpendicular to the surface, it acts straight down. Remember that the centripetal force is *not* a force that acts on the object, it is the *net force.*

KEY POINT :

For an object on the 'inside' of the circle at the top, we can write the equation of motion as follows. In this equation, *down* is taken as positive (remember to be consistent with your sign convention!)

$$F_{net} = F_c = W + N$$

$$\frac{mv^2}{r} = mg + N$$

$$\Leftrightarrow N_{(down)} = \frac{mv^2}{r} - mg$$

An interesting scenario, often used in questions, is when the object is travelling *slow* enough to lose contact with the circular path it's travelling on and literally fall off the tracks. To solve such scenarios, we must understand that the *normal force would be zero* as it leaves the surface because there is no more contact. Therefore, we would plug $N = 0$ into our equation and solve.

EXTENSION :

We can easily derive a shortcut formula for such a scenario, although (as always) I stress the necessity of knowing how to tackle questions from scratch without the need for shortcut formulas.

We know that from above:

$$N_{(down)} = \frac{mv^2}{r} - mg$$

Now plug in $N = 0$,

$$0 = \frac{mv^2}{r} - mg$$

$$\therefore v = \sqrt{gr}$$

Again, to find the minimum speed for an object to leave the surface we don't actually need to know its mass! And also note that the resulting formula is *exactly* the same as the shortcut formula for **Case 1**.

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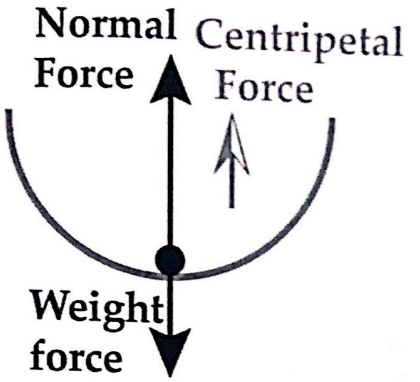
3.1.4.5 Solving Vertical Circles (Lowest Point)

The next scenario we consider is when the object is at the bottom of the circle. However, we will only consider the only case when the object is 'inside' the circle because the object is rarely ever on the outside, dangling from the bottom.



From the diagram, it's evident again that there are only *two* forces that act on the object: the normal force and the weight force. Remember that the centripetal force is *not* a force that acts on the object, it is the *net force.*

KEY POINT :

For an object on the 'inside' of the circle at the bottom, we can write the equation of motion as follows. In this equation, *up* is taken as positive (remember to be consistent with your sign convention!)

$$F_{net} = F_c = N - W$$

$$\frac{mv^2}{r} = N - mg$$

$$\Leftrightarrow N_{(up)} = \frac{mv^2}{r} + mg$$

3.1.5 Projectile Motion

The term 'Projectile Motion' is used to describe any object that is launched or dropped at a given velocity and allowed to travel unhindered. Unless stated otherwise, air resistance is neglected and the only force that acts on the object is gravity.

This means that we can use our constant acceleration formulas as the net force is gravity which is constant (if the object is close to the Earth's surface). To accurately include air resistance in our calculations would be beyond the scope of VCE Physics.

3.1.5.1 Methods of Solving

For almost all projectile motion questions, we split up the motion into the vertical and horizontal planes and solve respectively. As the time taken to travel in the vertical and horizontal directions are the same, we can use it for both sets of equations.

KEY POINT :

Using the following formulae which come from the *constant acceleration formulas*, we can substitute in information from the question to solve for the required unknowns.

Work in either the vertical or horizontal direction then use the time value to solve for the unknowns in the other direction.

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Initial launch:

$$\text{initial velocity} = u \quad \text{with angle} = \theta$$

$$u_V = u\sin(\theta)$$

$$u_H = u\cos(\theta)$$

Vertically, gravity is the only force, so $a = -g$ (since gravity acts down)

$$v_V = u_V - gt$$

$$x_V = u_V t - \frac{1}{2}gt^2$$

$$v_V^2 = u_V^2 - 2gx_V$$

$$x_V = \frac{1}{2}(v_V + u_V)\,t$$

Horizontally, no force acts so $a = 0$

$$v_H = u_H$$

$$x_H = u_H t$$

Remember that time is constant, so

$$t_V = t_H$$

Also recall that to find the final velocity from v_V and v_H:

$$v = \sqrt{v_V^2 + v_H^2}$$

$$\theta = \tan^{-1}\left(\frac{v_V}{v_H}\right)$$

3.1.6 Energy

Another way of describing motion is to consider the amount of energy an object has at a particular point in space and at a given time. We can use energy calculations to solve for quantities such as velocity. It is important to keep in mind that energy is a *scalar* quantity measured in Joules (J).

3.1.6.1 Kinetic Energy (KE)

KEY POINT :
Kinetic Energy is the energy due to motion. To change the kinetic energy of an object, an external unbalanced force must be applied.

Kinetic Energy is calculated as

$$E_k = KE = \frac{1}{2}mv^2$$

where KE is in J, m is in kg, and v is in m/s

3.1.6.2 Gravitational Potential Energy (GPE)

KEY POINT :
Gravitational Potential Energy is the energy stored in an object because of its position above some datum/reference point (usually taken as the Earth's surface). To change the gravitational potential energy, the height of the object must be changed.

Gravitational Potential Energy is calculated as

$$U_g = GPE = mgh$$

where GPE is in J, m is in kg, g is in m/s and h is in m

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3.1.6.3 Elastic Potential Energy (EPE)

Key Point :
A force must be applied in order to stretch or compress (*deform*) an object.

An ideal spring is an elastic material whose length changes *directly proportional* to the force applied.

Hooke's Law can be used to find the relationship between the force applied and the stretched length. Note that x is the *change* in length, sometimes written as Δx, not the final length.

$$F = -kx$$

where F is in N, k is in Nm^{-1} and x is in metres

Key Point :
Elastic Potential Energy, also known as *Strain Potential Energy* is the energy stored in an object because it has the ability to return from a deformed (stretched or compressed) state. To return to its neutral position, the external stretching or compressing force must be released.

For an *ideal spring* that obeys Hooke's Law, we can directly calculate the elastic potential energy given that we know the value of k. Note that x is the *change* in length, sometimes written as Δx, not the final length

$$U_e = EPE = \frac{1}{2}kx^2$$

where EPE is in J, N is in Nm^{-1} and x is in metres

3.1.7 Momentum

Key Point :
Momentum is "mass in motion". It is a *vector* quantity calculated as the product of velocity and mass of an object, and has the same direction as the velocity of the object. The greater the momentum of an object, the longer the time taken for it to come to a stop due to some external force.

Momentum has units of kg.m/s or N.s.

$$p = mv$$

...where p is in kg m/s, m is in kilograms, and v is in metres per second.

3.2 Einstein's Theory of Special Relativity

3.2.1 Einstein's two Postulates

Key Point :
Einstein's theory of Special Relativity is based on two postulates:

1. The laws of Physics are the same in all inertial reference frames.
2. The speed of light in a vacuum, c, is the same for all observers, regardless of any velocity the observer may have.

Keep in mind that an inertial frame of reference is one where the 'viewer' is at rest of at a constant velocity. For example, if you are sitting in a car that accelerates, your 'perspective' is not an inertial frame of reference.

Einstein's second postulate appears to contradict relativity in classical Physics. For example, if you are standing still and someone runs towards you at 10m/s, you see them running towards you at 10m/s. However, you are also running towards then at 10m/s, then they appear to be moving towards you at 20m/s.

However, when measuring the speed of light, even if you are travelling towards the source of light at 10m/s, the speed of light you measure is still c, not c + 10m/s.

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3.2.2 Proper Time and Length

Einstein's 'explanation' of his second postulate was that time and space are not absolute. For example, suppose there are two viewers, one on a train and one standing at a platform. If Bob on the train turns on a light in the train and it bounces off the roof and back, then it appear to him that it has travelled in a straight vertical line. However, for Anna standing on the platform, it appears that the light has travelled in a triangular path of greater distance.

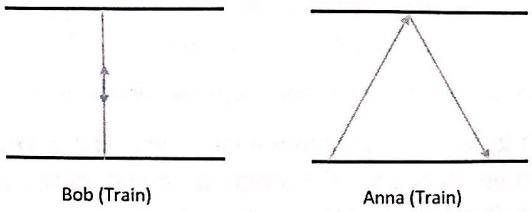

Figure 3.2: Non-simultaneity to show the difference in time and length

Thus, it is clear that if light travels at the same speed, either the time and/or distance travelled measured by Bob will be different than Anna's measurements. Proper time and length are defined by a rest frame of reference that is the same as the event being measured.

KEY POINT :
The time interval between two events that occur at the *same* place, e.g. both on a train or both on a platform, measured by an observer in that reference frame is called the proper time, t_0.

The length of an object measured in a reference frame in which it is at *rest* is called its proper length, L_0.

3.2.3 Time Dilation and Length Contraction

KEY POINT :
The Lorentz Factor is denoted by γ. For an observer moving at a relative speed v to the event, the Lorentz Factor is given by:

$$\gamma = \frac{1}{\sqrt{1-\left(\frac{v^2}{c^2}\right)}}$$

When the relative speed increases, time appears to travel more slowly. In the previous train example, if the train travels faster, Anna will record an even longer time for the light to reflect back from the ceiling to the ground.

KEY POINT :
Time dilation is the increase in the measured time, found by:

$$t = \gamma t_0$$

While time appears to travel more slowly, the distance measured appears to be shorter.

KEY POINT :
Length contraction is the decrease in the measured distance, found by:

$$L = \frac{L_0}{\gamma}$$

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3.2.4 Application: Muons

Certain muons have an average lifetime of 2.2 μs measured at rest. However, when travelling at a speed of 0.9994c they lifetime is suddenly much larger, 63.4 μs indicating a Lorentz factor of 28.9.

3.3 Relationship Between Force, Energy and Mass

3.3.1 Momentum-Impulse

3.3.1.1 Impulse

KEY POINT :
Impulse is the total effect of a force over the time the force acts. It is a *vector* quantity calculated as the product of the force acting on an object and the length of time over which the force acts. Impulse has the same direction as the velocity of the object.

Impulse has the units of N/s, which is the same as the units for momentum, kg.m/s.

$$I = Ft$$

where I is in Ns, F is in N and t is in s

...where I is in Newtons per second, F is in Newtons, and t is in seconds

If the force is *not constant*, the impulse is calculated by finding the area under a force-time graph. Or, the average force can be used, meaning that $I = F_{ave}t$.

3.3.1.2 Link Between Momentum and Impulse

KEY POINT :
In a collision, an object experiences a force for a specific amount of time that results in a change in momentum, causing the object to slow down or speed up. This change in momentum is *equal* to the impulse applied to the object.

This can be expressed mathematically as

$$I = p_{final} - p_{initial} = \Delta p$$

.

EXTENSION :
We can actually prove this using Newton's second law.

Newton's Second Law:

$$F = ma$$

Use constant acceleration formulae:

$$F = m\left(\frac{v-u}{t}\right)$$

$$Ft = mv - mu$$

$$I = p_{final} - p_{intial}$$

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3.3.2 Work Done

KEY POINT :
Work in Physics relates to the concept that energy is transferred or transformed when a force acts on a body *causing* a change in displacement. To indicate that work has been done, simply state that: **work has been done on the object by the external force.**

There are cases when a force is applied but no work is done by an external force. For example, if the final displacement is zero, then by definition no work was done as there was *no change* in displacement.

If the force acts at right angles to the displacement, then no work is done because the force did not *cause* the change in displacement.

To calculate the work done on an object, we can determine the change in energy (e.g. kinetic energy) or the displacement multiplied by the force component acting parallel to the displacement.

$$W = \Delta E = x \cdot F\cos(\theta)$$

where W is in J, E is in J, x is in m, F is in N and θ is in degrees

If the force is non-constant, then the work done is found by estimating the area under a force-time graph.

3.3.3 Conservation of Energy

KEY POINT :
"Energy cannot be created or destroyed, it can only be changed from one form to another."

Note that is is only in classical Physics, i.e. no special relativity!

Mechanical Energy the is total of all *kinetic and potential energies*. In a *frictionless system*, the mechanical energy is constant.

$$KE + GPE + EPE = const.$$

$$\frac{1}{2}mv^2 + mgh + \frac{1}{2}kx^2 = const.$$

In a system where friction is not negligible and cannot be neglected, the total mechanical energy will not stay constant. Energy is *not* destroyed, it is transformed into less useful forms.

KEY POINT :
"Energy loss" is when energy is degraded into less useful forms such as heat, sound, or permanent deformation (such as a permanently crushed Coke can).

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3.3.4 Mass-Energy

In addition to non-absolute time and length, Einstein's relatively indicates that mass is also not absolute.

KEY POINT :

The rest mass of an object is the mass measured at rest, denoted by m_0.

When moving at a speed v, the mass of the object is given by

$$m = \gamma m_0$$

Now, since $\gamma \geq 1$ and increases as v increases, as the velocity of an object approaches the speed of light its mass approaches infinity. This means that an infinite amount of energy would be required to continue accelerating the mass.

KEY POINT :

Mass and energy are equivalent giving rise to Einstein's most famous formula,

$$E = mc^2$$

For example, if one gram of matter is completely converted to energy, the amount of energy released would be 9×10^{13}J - a tremendous amount!

KEY POINT :

As mass and energy are no longer considered separately, it is the total mass-energy of a system that is conserved.

$$E = E_0 + KE$$

...where E is the total energy, E_0 is the rest mass energy $E_0 = m_0c^2$ and KE is the kinetic energy.

Suppose an object is travelling at very low speeds, then

$$KE = E - E_0 = mc^2 - m_0c^2 = \gamma m_0c^2 - m_0c^2$$
$$\Rightarrow KE = (\gamma - 1)\, m_0c^2$$

For low speeds of v, this is very close to the kinetic energy in our classical physics,

$$KE = \frac{1}{2}mv^2$$

3.3.5 Application: Fusion in the Sun

A fission reactor, such as that in a nuclear power plant, involves 'splitting' a larger atom such as uranium into two smaller ones. However, in a fusion reactor the opposite occurs – two smaller atoms are combined into a single larger atom. Both of these processes create large amounts of energy by converting some mass into energy according to $E = mc^2$.

The sun like any star begin as a large cloud of gas that contracted under its own gravity. At its core hydrogen atoms (the majority of the gas) are subject to immense pressures, about 100 billion times greater than the air pressure on the surface of the Earth and temperatures. This is enough to force two proton together enough though they repel each other.

Four hydrogen atoms (four protons) are required for fusion which produce a Helium particle (two protons and two neutrons), two positrons and two neutrinos. The mass "lost", i.e. the difference in mass between a Helium particle and four Hydrogen particles, is approximately 4.77×10^{-29}. Although this is a tiny mass, it equates to 4.29×10^{-12}J per reaction. There are around 9×10^{37} reactions per second occurring in our sun which equates to a near-ridiculous amount of energy being produced!

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Part II

Unit 4: How can two contradictory models explain both light and matter?

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Area of Study 1

How can waves explain the behaviour of light?

1.1 Properties of Mechanical Waves

1.1.1 Energy Transfer of Waves

KEY POINT :
Waves are the transfer of energy from one place to another without the *net* transfer of matter.

Mechanical waves such as sound require a medium such as air to travel through.

1.1.2 Measurements of Waves

- **Period (T):** is the time taken for a complete cycle measured in seconds (s).
- **Frequency (f):** is the number of cycles in one second measured in Hertz (Hz).
  - These two quantities are related by: $f = \frac{1}{T}$.
- **Wavelength (λ):** is the length of a single cycle measured in metres, m, or the distance a wave travels during one period (T).
- **Velocity (v)** The speed at which a wave travels measured in metres per second, ms^{-1}. For mechanical waves, such as sound, this can change depending on the medium, such as air or water.
- **Wave Equation** Relates the frequency, wavelength and velocity as $v = f\lambda$.
- **Amplitude** The larger the amplitude of the wave, the greater the energy (often referred to as intensity).

1.1.3 Interference of Waves

KEY POINT :
Interference of waves occurs when two waves meet causing a resultant wave that is the net effect of the individual waves.

Constructive interference occurs when two waves meet with displacements in the same direction. For example, if two wave peaks meet each other, the net result will be an even larger peak, i.e. the waves have increased in strength.

Destructive interference occurs when two waves meet with displacements in different directions. For example, if a wave peak meets a wave trough then the net result will be zero, i.e. the waves have cancelled each other out.

1.1.4 Application: Doppler Effect

The classic example of the Doppler Effect is a police car or fire truck driving along a street with its siren blaring. When the siren is travelling towards you, it has a higher pitch and when it travels away it appears to have a lower pitch.

KEY POINT :
The Doppler Effect is produced when the source of a wave moves with respect to the observer causing an apparent shift in frequency for the observer.

It is important to note that the frequency does not *actually* change, it merely appears to for the observer.

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So, for the police officer in the car, the siren stays at the same pitch. This is because the siren is playing the same sequence of notes over and over.

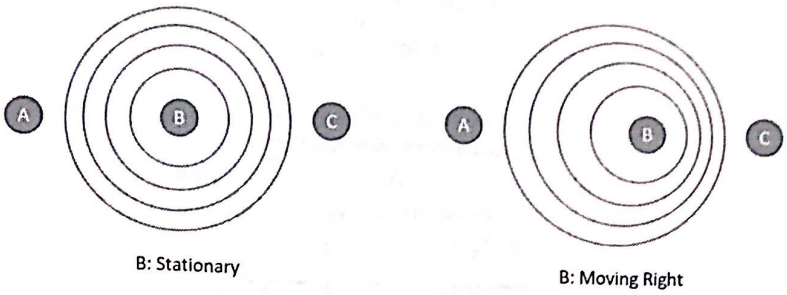

Figure 1.1: Doppler Effect for Various Observers

In Figure 1.1 on page 39 we can see that when the source is stationary A, B and C will all experience the same frequency, the same as the one produced at the source.

However, when moving towards C we can see that the wave-fronts (shown as lines) are closer together indicating a higher frequency. Conversely, as the source of the wave is moving away from A, the wave-fronts are further apart indicating a lower frequency. Only B will experience the true frequency as B is 'at the source'.

1.1.5 Resonance

Every system, such as a bridge, has one or more frequencies at which it will vibrate *naturally* if disturbed. For example, when you pluck a guitar string it will vibrate at a given natural frequency. If you try to force the guitar string to vibrate at a different frequency, the vibration will quickly die off.

KEY POINT :
Resonance occurs when an external force vibrates a system at a frequency that exactly matches one of the natural frequencies.

- The system absorbs energy from the driving force (vibrations do not die down quickly).
- The system vibrates with a large amplitude.

1.1.6 Standing Waves

KEY POINT :
A standing wave is a specific type of wave pattern where a wave is reflected and interferes with itself constructively. At various points the standing wave will have zero displacement (nodes) and maximum displacement at others (anti-nodes).

Standing waves can only be formed at very specific frequency. The reason for this is because when the wave is reflected back the driving wave at the other end must be exactly in phase, i.e. a peak meets a peak and a trough meets a trough.

The common examples of standing waves are: strings tied at both ends, tubes with both ends open, or a tub with one end open.

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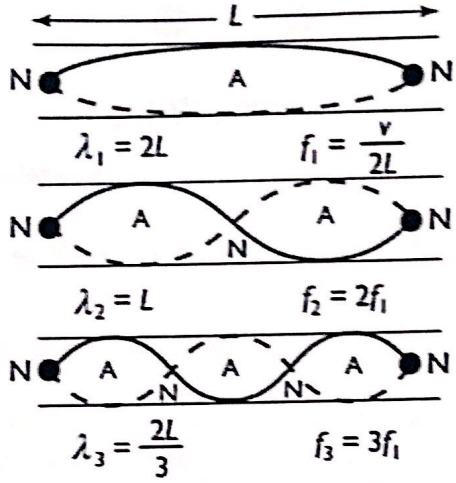
1.1.6.1 Strings Tied at Both Ends

To imagine a standing wave on a string, imagine plucking a guitar string. Because both ends are fixed, this means that the vibration (displacement) of the string must occur in the 'middle' of the string.

There are many possibilities for the standing wave, for example:



If we wanted to talk about the general formulae for the wavelength and frequency, we could find them as:

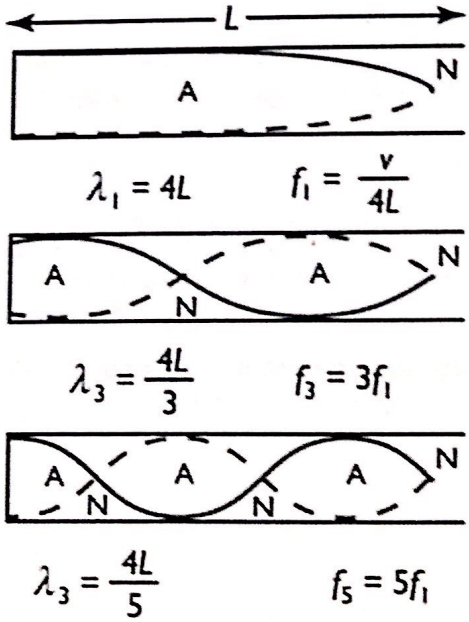
$$\lambda_n = \frac{\lambda_1}{n} = \frac{2L}{n}, \qquad f_n = nf_1 = \frac{nv}{2L} \qquad \text{where } n = 1, 2, 3, \ldots$$

1.1.6.2 Tubes Open at Both Ends

A tube open at both ends acts very similarly to a string tied at both ends. The reason for this is because at the open ends there must be zero pressure (i.e. zero displacement). So, the shape and formulae for these situations are the exact same as a string fixed at both ends.

1.1.6.3 Tubes with one End Closed

A tube with one end closed is quite different to a tube with both ends open. The reason for this is as the standing wave reaches the closed end of the tube, it will constructively interfere with itself causes an anti-node (maximum pressure or displacement).



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1.1.7 Diffraction

KEY POINT :
When light passes through a slit that has a width w that is similar to its wavelength λ, the light spreads out forming a diffraction pattern.

The spacing is proportional to $\frac{\lambda}{w}$, so as the wavelength increases the diffraction expands and when the slit width increases the diffraction contracts. Another way to express this is:

$$diffraction \propto \frac{wavelength}{slit\ size} = \frac{\lambda}{w}$$

- $w < \lambda$ means significant diffraction occurs.
- $\lambda < w$ means less diffraction occurs.

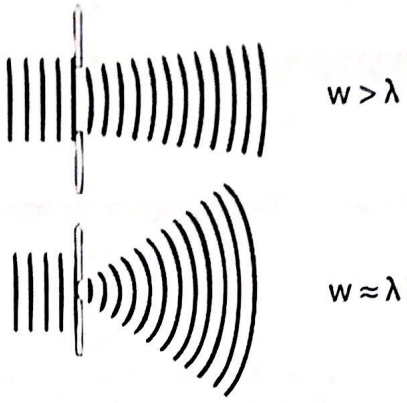

Figure 1.2: Diffraction effects for different slit widths

Figure 1.2 on page 41 shows the relative diffraction for different slit widths. Similarly, if there is an obstruction of width w, the wave will also diffract around the obstruction.

1.2 Light as a Wave

1.2.1 Electromagnetic Waves

Maxwell predicted that an accelerated electric charge could produce an electromagnetic wave. These waves were oscillating perpendicular electric and magnetic fields that don't require a medium (e.g. sound requires air to travel) meaning they can travel through space.

In space, light and other electromagnetic waves travel at c, the speed of light.

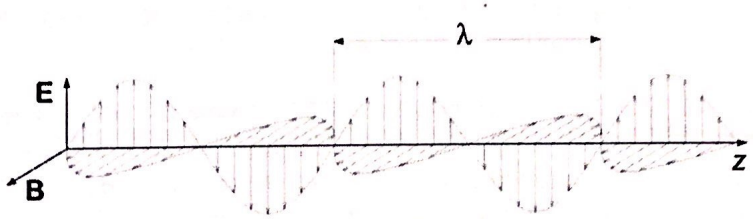

Figure 1.3: Electromagnetic Wave

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1.2.2 Electromagnetic Waves Spectrum

The electromagnetic spectrum is a whole range of electromagnetic waves produced by accelerating an electric charge. Light is only a small part of the electromagnetic spectrum, classified as visible light.

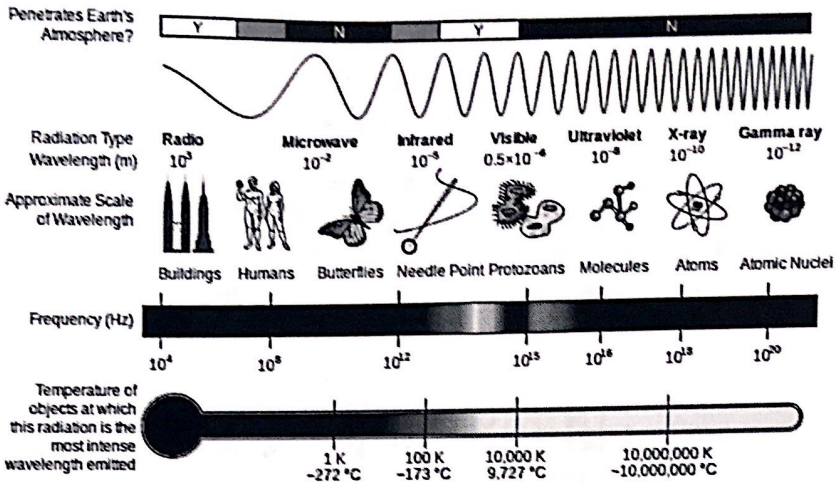

Figure 1.4: Electromagnetic Spectrum

1.2.3 Polarisation of Light

As was touched on earlier, light is an electromagnetic wave that consists of perpendicular electric and magnetic fields shown in Figure 1.3 on page 41. When there are vibrations in more than one direction (plane), light is classified as being unpolarised. Light emitted from the sun is unpolarised.

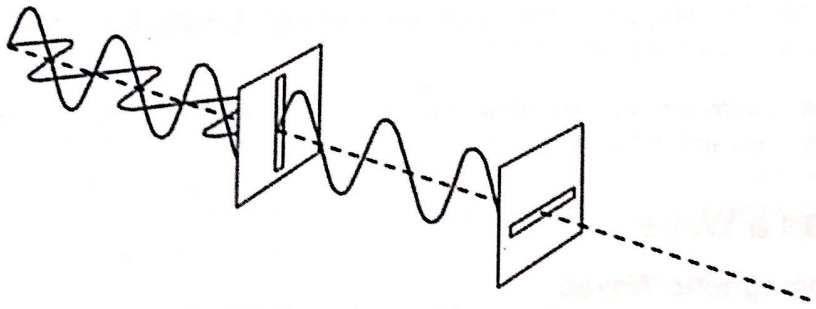

Figure 1.5: Light passing through polaroid filters

KEY POINT :

Polarisation of light is when light passes though a polariod filter which blocks out light waves in one direction.

A polaroid filter works due to a complex chemical reaction. These are not only popular in sunglasses because they effectively filter out half of light, but also at 3D cinemas. One of the lens in your 3D glasses will have a horizontal polaroid filter and the other will have a vertical filter. Because they are are filter out different sections of light, you will actually see a different image in each eye! The only downside is you can't move you head too much, e.g. if you rotate your head 90 degrees the lenses will polarise the light in the wrong direction meaning the image meant for your right eye will be seen by your left eye and vice versa.

Light can also be partially polarised when it reflects off a non-metallic surface. The amount of polarisation depends on the angle of incidence and the type of material. If the extend of polarisation is large, this can cause significant glare to a viewer.

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1.2.4 Refraction

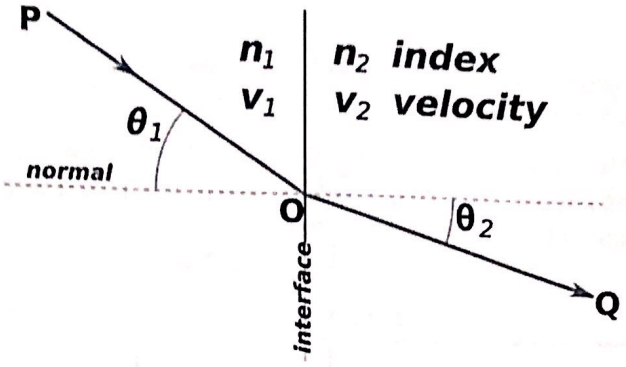

Figure 1.6: Refraction of light as it passes between two mediums

When a wave passes from one medium to another, it will refract meaning it changes direction. The reason for this is because the frequency of the wave does not change, but the speed does. It is possible to use conservation of energy and conservation of momentum to deduce the change, but this is not required for VCE Physics.

KEY POINT :
Snell's law can be used to describe refraction:

$$\frac{\sin(\theta_1)}{\sin(\theta_2)} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

...where θ is the angle measured from the *perpendicular* to the surface, v is the velocity of the wave in the medium, and n is the refractive index. The refractive index for light is found as:

$$n = \frac{c}{v} = \frac{\text{speed of light in a vacuum}}{\text{speed of light in a medium}}$$

Refraction explains why straws appear to 'bend' when they are placed in glasses of water and the appearance of rainbows which is due to different wavelengths of light being refracted differently.

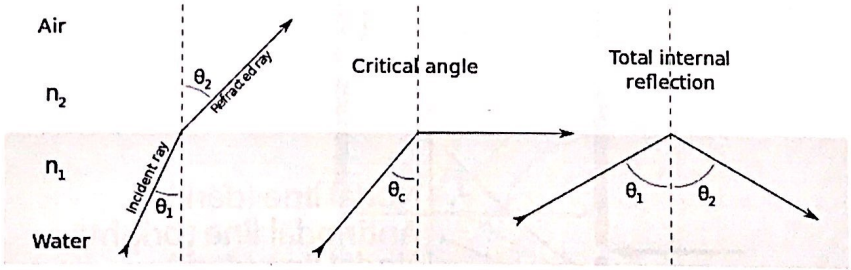

Figure 1.7: Critical Angle of Refraction

The critical angle is the minimum angle for total internal reflection. This can be found by setting $\theta_2 = 90°$ which gives:

$$\theta_C = \sin^{-1}\left(\frac{1}{n_1}\right)$$

This is important for many applications, especially fibre optic cables. In fibre optic cables light signals are passed through the cable and are not 'lost' a long the way because total internal reflection occurs rather than refraction. That is why it is possible to bend fibre optic cables without light 'escaping' at the bend.

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1.2.5 Young's Double Slit Experiment

KEY POINT :
Young's double slit experiment set-up:

- Monochromatic light with a wavelength λ shines on two slits that are a distance d apart.
  - The light passes through the slits and falls on a screen that is a distance L away.

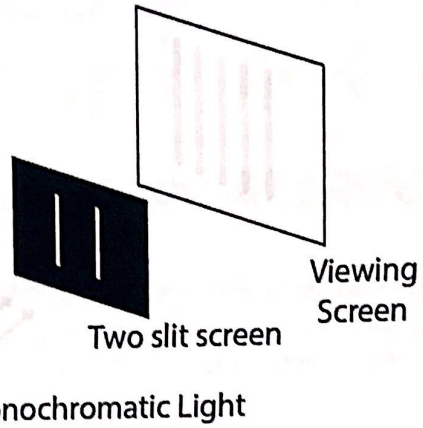

Figure 1.8: Setup of Young's double slit experiment

Monochromatic light is used as it only has a *single* wavelength, in other words the light is all the 'same' and cannot be broken up into a rainbow spectrum like white light. This is so that we know our results are due to diffraction and interference (superposition) of waves and not due to other effects.

KEY POINT :
Young's double slit experiment results:

- An interference pattern is observed on the screen.
- On the screen there are bands of light (anti-nodal) and dark (nodal) lines.
- The fringes (bands) produced are evenly spaced.
- The intensity of light is greatest at the centre and decreases as the bands get further from the centre.

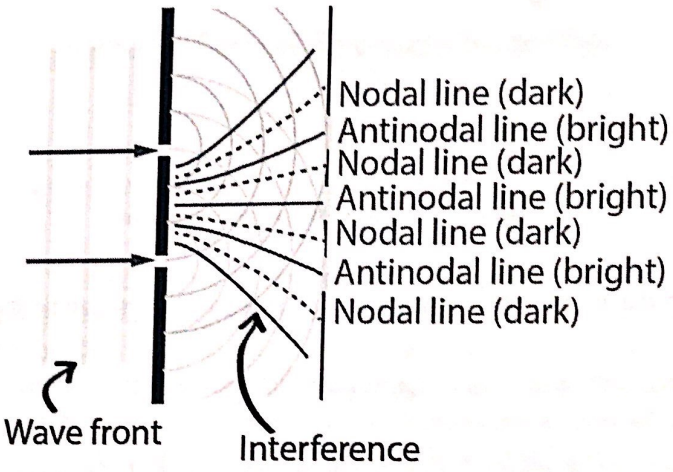

Figure 1.9: Interference pattern of Young's double slit experiment

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Figure 1.9 on page 44 shows the interference of the waves as they travel from the slits to the screen. The term 'node' is used for dark bands (fringes), and 'antinode' is used for light bands (fringes). The reason for these is because of *superposition*.

The wave front represents the crest of a wave, and the gap represents a trough. When two lines (crests) meet, they 'add together' (constructive interference) to form a bigger crest (bright line). When two gaps (trough) meet, they also 'add together' (constructive interference) to form a bigger trough (also a bright line). So at these points where line meets line or trough meets trough, we get a 'bigger', brighter wave.

When a line (crest) meets a gap (trough), they will 'cancel each other out' (destructive interference). So there means there is effectively no wave at that point, i.e. a dark patch.

This is why there is a striped pattern of bright and dark bands.

EXTENSION :

When changes are made to the experiment, the resulting fringes are changed. A useful formula is:

$$x = \frac{\lambda L}{d}$$

...where x is the distance between two light bands (or the distance between two dark bands), λ is the wavelength of light used, L is the distance between the slits and the screen, and d is the distance between the slits. This is shown in Figure 1.10 on page 45.

A fairly common question that is asked on the exam is, what happens if the wavelength is increased. You could think it out in terms of geometry, but it is *much* simpler to look at the formula and think: λ increases so x increases too. These questions are generally one mark or multiple choice, so examiners aren't too fussed on how you get your answer, just as long as it is right!

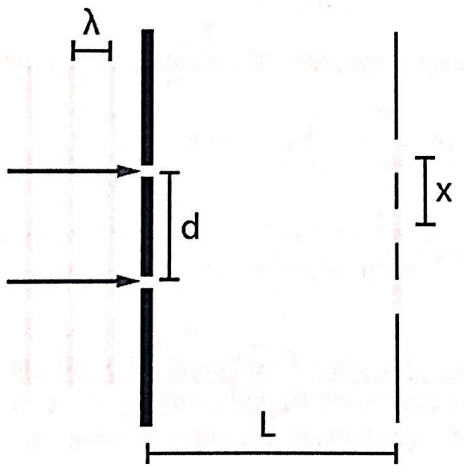

Figure 1.10: Important values in Young's double slit experiment

Example 1.1

A student performs his own Young's Double Slit Experiment because he doesn't believe that light is a wave. What result would the particle model for light predict? Why is this incorrect?

Suppose light were a stream of tiny tennis balls. When they reach the slits, they either pass through the slits or hit the surrounding wall and do not pass through. But, they would *not* diffract as they pass through the slits. So, they continue to travel in a relatively straight line until they hit the screen and form *two* bright bands. This prediction is wrong because light *does* diffract as it pass through the slits and forms two distinct waves that interfere with each other constructively and destructively at different points. This is seen on the screen as a diffraction pattern of light and dark bands or fringes.

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1.2.5.1 Path Difference and Interference

How can we mathematically explain interference?

Key Point :

At any point on the screen (P), a wave from slit one (S_1) will have travelled a distance S_1P and a wave from slit two (S_2) will have travelled a distance S_2P. The difference in the distance travelled is the path difference,

$$pd = |S_1P - S_2P|$$

We can measure path difference in metres, but is usually measured in wavelengths to determine if the spot P is bright or dark.

At the centre point, the distance from slit one (S_1) to the centre (C) is the same as the distance from slit two (S_2) to the centre (C). This means that when light from each of the slits reaches C, they are in the same phases as they have travelled the same distance: $S_1C = S_2C$ and $pd = 0$.

Now, the two waves have travelled the same distance which means that they would interfere *constructively* as they are in the same phase, i.e. a crest meets a crest and a trough meets and trough. So, the centre band is *always* bright.

Key Point :

Constructive interference occurs when the path difference is a multiple of λ, that is

$$pd = n\lambda \qquad \text{where } n = 1, 2, 3, \ldots$$

This is because at point P, even though the waves have travelled a different lengths, the waves arrive in the *same* phase, i.e. a trough meets a trough and a crest meets a crest.

Key Point :

Destructive interference occurs when the path difference is an odd multiple of λ, that is

$$pd = \left(n - \frac{1}{2}\right)\lambda \qquad \text{where } n = 1, 2, 3, \ldots$$

This is because at point P, the waves have travelled a different lengths and arrive in the *opposite* phase, i.e. a trough meets a crest and a crest meets a trough.

Example 1.2

Monochromatic light of wavelength λ is used for a Young's Double Slit Experiment. The path difference from the centre bright band to the second dark band on the right is 675 nm. Find the path difference from the centre bright band to the third bright band on the left in nanometres.

Begin by noting a dark band means *destructive* interference. So, our pd is given by:

$$675\,\text{nm} = pd = |S_1D_2 - S_2D_2| = \left(n - \frac{1}{2}\right)\lambda \qquad \text{where } n = 1, 2, 3, \ldots$$

Now, since it is our second dark band, $n = 2$

$$675 = 1\frac{1}{2}\lambda \rightarrow \lambda = 450\,\text{nm}$$

Now, the bright band meens *constructive* interference. So, our pd is given by:

$$pd = |S_1B_3 - S_2B_3| = n\lambda \qquad \text{where } n = 1, 2, 3, \ldots$$

Now, since it is our thrid bright band, $n = 3$

$$pd = 3\lambda \rightarrow pd = 450 \times 3 = 1350\,\text{nm}$$

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Area of Study 2

How are light and matter similar?

2.1 Behaviour of Light

2.1.1 Photoelectric Effect

2.1.1.1 Experiment Set-up

It was discovered that when light shines on a metal plate, electrons are emitted from the surface. These electrons are called *photoelectrons*, the 'photo' prefix refers to light being the cause of these electrons being emitted. When a steady flow of electrons is present, it is called a *photocurrent*. A photocurrent flows from the cathode to the anode in a circuit.

KEY POINT :

The photoelectric effect experiment consisted of:

- A metal placed at the cathode,
- A monochromatic (single wavelength) light source that shone onto the cathode,
- An ammeter to detect photocurrent, and
- A variable voltage that could provide current in the same or opposite direction to the photocurrent (if the circuit was closed).
  - Forward potential: the variable voltage would make the anode positive to help the photoelectrons move from the cathode to the anode, or
  - Reverse potential: the variable voltage would make the anode negative to prevent photoelectrons from the cathode reaching the anode.

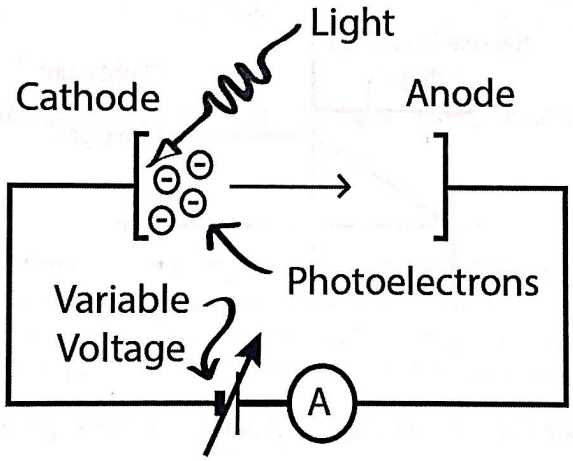

Figure 2.1: Basic operation of the Photoelectric effect experiment

Figure 2.1 on page 47 shows the set-up of the photoelectric effect experiment. We can see that at first the electrons are clustered at the cathode, and will travel to the anode to complete the circuit. Since we know that opposites repel, if we make the cathode negative it will repel the electrons and 'help' then cross to the waiting anode. That is forward bias. So, if we make the anode negative, it will repel the electrons and try to 'stop' them from reaching it. This is reverse bias, and is often called 'retarding voltage' where 'retarding' refers to the slowing/stopping of electrons reaching the anode.

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2.1.1.2 Experiment Results

KEY POINT :
The important results of the photoelectric experiment include:

- For each metal used at the cathode, there was a unique *threshold frequency*, f_0. If the frequency of light used was below this value, no photoelectrons were emitted.
- There was no delay between light shining on the cathode and photoelectrons being produced.
- When the voltage was increasing in reverse potential (increasing retarding voltage), eventually the photocurrent would drop to zero at the stopping voltage, V_s.
- If the intensity of light (brightness) increased, the stopping voltage *did not* change.
- If the intensity of light increased, the photocurrent increased proportionally to the intensity.
- If a larger frequency of light was used, a larger stopping voltage was needed.
- If a larger frequency of light was used, the maximum photocurrent *did not* change.

What would the wave model have predicted?

- There *should* be a time delay before any electron has enough energy to escape. This is because waves are continuous, so it has to absorb a certain 'length' of a wave before it has enough energy to escape... wrong!
- The energy of the electrons *should* depend on the intensity of light. So, if a bright light is used, a *larger* stopping voltage should be needed to stop these energetic photons... wrong!
- The frequency should have *no effect* on the emission of photons... wrong!
- The energy of the electrons *should not* depend on the frequency of light used. So the frequency of light should have *no* effect on the stopping voltage... wrong!

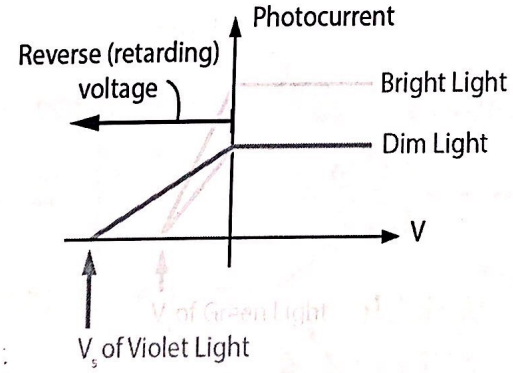

Figure 2.2: Results of the Photoelectric effect experiment

Figure 2.1 on page 47 shows some of the results for a particular experiment. The *key* things to note have already been listed, but this is how the information is graphically shown.

Example 2.1
It is known that for a particular metal, when mono-chromic red light is used, the stopping voltage required is 3.4 eV. If the intensity (brightness) of light is doubled, what change would occur according to the wave model? How is it different from the actual results?

The wave model would predict that the electrons would have a greater amount of kinetic energy, as the energy provided by a wave is proportional to its intensity. Since our photoelectrons have more energy, a larger stopping voltage will be needed. This idea was disproved as the stopping voltage did *not* change. So, the kinetic energy of the electron did not depend on the intensity of light. The actual change was that the photocurrent increased, since there are more particles bombarding the cathode releasing more electrons; creating a larger current.

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2.1.1.3 Photon Model of Light

Now let us expand our model. We often hear of light being referred to as 'photons', so let's look at what these actually are!

Light is emitted and absorbed in tiny discrete packets of electromagnetic energy called *photons* or *quanta*.

KEY POINT :
The energy of a photon is given by:

$$E_{photon} = hf = \frac{hc}{\lambda}$$

...where E_{photon} is the energy of the photon (measured in Joules, J), h is Planck's constant which is equal to 6.63×10^{-34}J s (Joule seconds), f is the frequency (measured in hertz, Hz), c is the speed of light which is equal to 3.00×10^{8} m s^{-1} (metres per second), and λ is the wavelength (measured in metres, m).

Alternatively, we can find E_{photon} in terms of electron volts, eV. Here, E_{photon} is the energy of the photon (measured in electron volts, eV), h is Planck's constant which is equal to 4.14×10^{-15}eV s (electron volt seconds). The total energy of a light beam can be found by,

$$\text{energy of light beam} = \text{number of photons} \times \text{photon energy}$$

So now that we know that light seems to travel in little packets of energy, how does this explain our findings in the photoelectric effect experiment?

2.1.1.4 Maximum Kinetic Energy of Electrons

The maximum kinetic energy for an electron can be found from the photocurrent vs. voltage graph. The stopping voltage is the voltage needed to stop even the most energetic electrons, so it is a direct measure of the maximum kinetic energy.

$$KE_{max} = V_s \text{ (in eV)} \quad \text{or} \quad KE_{max} = e \times V_s \ (V_s \text{ in V})$$

KEY POINT :
Different electrons are bound by different energies in the cathode metal. The smallest amount of energy required to free one of these electrons is called the work function, W. Each metal as its own unique work function. When light at the threshold frequency f_0 is shone onto the cathode, photons have just enough energy to free electrons. In other words, the energy of the photon is equal to the work function:

$$E_p = hf_0 = W$$

When light with a frequency *less than* the threshold frequency is shone onto the cathode, none of the photons have enough energy to free electrons. In other words, the energy of the photon is less than the work function:

$$f < f_0, \quad E_p < W$$

When light with a frequency *greater than* the threshold frequency is shone onto the cathode, photons have more than enough energy to free electrons. The 'extra' energy from the absorbed photon is converted into the kinetic energy of the electron. That is:

$$KE_{max} = E_p - W = hf - W$$

$$\text{Since } W = hf_0, \quad KE_{max} = hf - hf_0 = h(f - f_0)$$

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So, there are two ways to find the kinetic energy of emitted electrons, either through the stopping voltage or the work function: $KE_{max} = V_S$ (in eV) $= hf - W$

Figure 2.3 on page 50 shows us the relation between the work function and the frequency. We can further analyse this graph by looking at it as a linear model.

KEY POINT :
The work function, kinetic energy, threshold frequency and Plank's constant can all be graphically related.

The equation $KE_{max} = hf - W$ resembles a straight line equation, $y = mx + c$. So we can say that:

- KE_{max} is the vertical axis and f is the horizontal axis.
- $-W$ (work function) is the 'y-intercept' and f_0 (threshold frequency) is the 'x-intercept'.
- h is the gradient of the line and remains constant even for different W and f_0 values of metals.

However, care should be taken when drawing the line below the horizontal axis. Since it is physically impossible to have a negative kinetic energy, we draw this section as a *dotted* line.

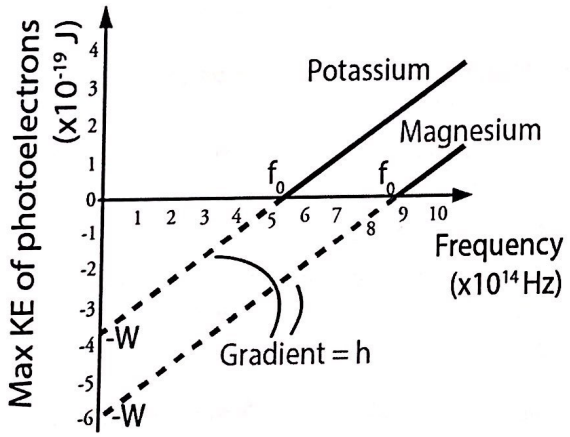

Figure 2.3: Relation of kinetic energy, frequency, work function, threshold frequency, and Plank's constant.

2.2 Matter as Particles or Waves

2.2.1 Diffraction Patterns of Photons and Electrons

An important implication is that since matter has a wavelength as well, it can also cause interference patterns.

A typical example used is that electrons passing through powdered metal foil can cause a diffraction pattern due to scattering and constructive interference between electron waves.

A beam of X-rays can also be shown on the same foil and it will also cause a diffraction pattern due to constructive and destructive interference.

KEY POINT :
The two patterns will *only* be the same if the de Broglie wavelength of the electron is the same as the wavelength of the X-ray. This is because the diffraction is related to $\frac{\lambda}{w}$. Since w does no change, the wavelength for both electrons and X-rays are the same. However, the electron and photons do *not* have the same speed or energy.

- The X-ray photon has zero mass and a fixed speed of c. It will have an energy of $E_p = \frac{hc}{\lambda}$
- The electron has a mass m and a speed given by $v = \frac{p}{m}$. It will have an energy of $KE = \frac{1}{2}mv^2$

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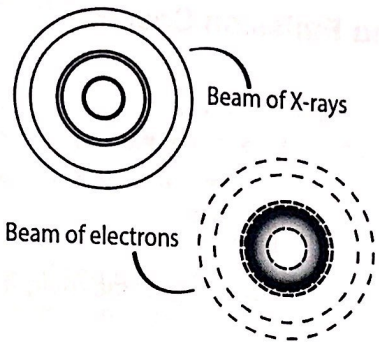

Figure 2.4: Diffraction results for X-rays and electrons

2.2.2 De Broglie Wavelength of Matter

If light can exhibit particle-like behaviour *and* wave-light behaviour, what about matter? Matter can also have wave-light behaviour!

KEY POINT :
Any moving particle of matter has an associated wave, known as the de Broglie wavelength which is given by:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

...where λ is the de Broglie wavelength (measured in metres, m), h is Plank's constant which is equal to 6.63×10^{-34} Js, p is the momentum of the particle (measured in kg ms^{-1}), m is the mass of the particle (measured in kg) and v is the velocity (measured in ms^{-1}).

Example 2.2
Find the de Broglie wavelength of a 150 *g ball travelling at* 20 ms^{-1}.

The wavelength is found by

$$\lambda = \frac{h}{p} = \frac{h}{mv} \rightarrow \lambda = \frac{6.63 \times 10^{-34}}{0.15 \times 20} = 2.21 \times 10^{34}\ \text{m}$$

This is a *tiny* value, which is why we don't notice wave-like behaviour in normal every day life. For example for diffraction to occur, $\frac{\lambda}{w} \approx 1$...so we would need the ball to fit in a slit the with of 2.21×10^{34} m!

2.3 Similarities of Light and Matter

2.3.1 Momentum

One of the final things to finish off our photon model, is to explain the concept of momentum of photons. It is indeed possible to find the momentum, even though photons are thought to have zero mass while travelling at the speed of light.

KEY POINT :
Photons also have a momentum, which is given by:

$$p = \frac{hf}{c} = \frac{h}{\lambda}$$

...where p is the momentum (measured in kg ms^{-1}), h is Plank's constant which is equal to 6.63×10^{-34} Js, f is the frequency (measured in hertz, Hz), c is the speed of light which is equal to 3.00×10^{8} ms^{-1} and λ is the wavelength (measured in metres, m).

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2.3.2 Atomic Absorption and Emission Spectra

KEY POINT :
When returning to their ground state from an excited state, atoms emit light. The light emitted consists of only a few bands of light forming what is known as a line emission spectrum. Each element has a unique spectrum which can be used for identification.

Each band corresponds to a specific frequency of light, meaning it is possible to find the energy of the emitted photon as $E_p = hf = \frac{hc}{\lambda}$.

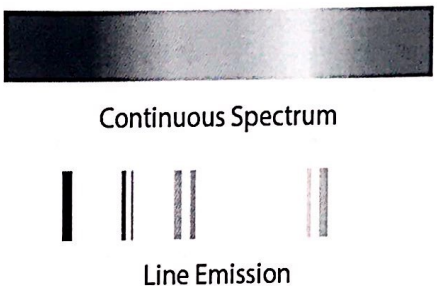

Figure 2.5: Example of a line emission spectra

KEY POINT :
When white light is passed through atoms, certain frequencies of light are absorbed. The light that exists is continuous expect for these places, and the spectrum is called a line absorption spectrum. Again, this spectrum is unique to each element.

Each band corresponds to a specific frequency of light, meaning it is possible to find the energy of the emitted photon as $E_p = hf = \frac{hc}{\lambda}$.

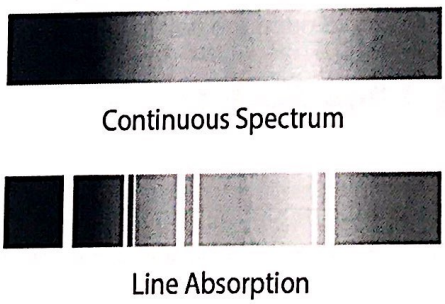

Figure 2.6: Example of a line absorption spectra

For any given element, for each dark band in the line absorption spectrum there is a corresponding light band in the emission spectrum. This means that atoms can only absorb photons with energies *exactly* equal to energies of photons that are emitted.

When a photon carrying the 'right' amount of energy hits an atom, an electron is knocked to a higher orbit (absorption) and when it returns to its original original orbit it emits an electron (emission).

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2.3.3 Quantised States of Atoms

2.3.3.1 Bohr's Model

The reason for these emission and absorption spectra is that there are specific energy levels that exist for an atom. Bohr's model is used to explain this in much more detail.

Bohr's model states that:

- An electron moves in a circular orbit around the nucleus (the electrostatic attraction of the nuclear (+) and the electron (-) is the source of centripetal force).
- There are only a certain number of allowable orbits at different distance from the nucleus which are called $n = 1, 2, 3, \ldots$
- Electrons do not emit energy (photons) when they are in one of these allowable orbits and ordinarily occupy the lowest orbit available (ground state).
- A photon absorbed has *exactly* the same energy as the increase (change) in energy of an electron in its current orbit jumping to a higher orbit.
- A photon emitted has *exactly* the same energy as the decrease (change) in energy of an electron in a higher orbit falling to a lower orbit.

2.3.3.2 Absorption and Emission of Electrons

So what happens to the energy that is lost when an electron drops to a lower orbit, or the energy gained when an electron jumps to a higher energy level? We know that energy cannot be created or destroyed (energy is conserved), so there must be something emitted or absorbed... a photon!

KEY POINT :
If an electron in an atom moves between an orbit m and n then the energy of the photon absorbed or emitted can be found by:

$$E_p = hf = E_m - E_n$$

...where E_m is the energy in orbit m, and E_n is the energy in orbit n.

Figure 2.7 on page 53 shows us a simple energy level diagram. On the left side, we have the vertical axis for energy. On the right side, we have the energy level name, $n = 1, 2, 3 \ldots$. The ground state is the lowest energy level, in other words where $E = 0$. This is also referred to as the $n = 1$ energy level.

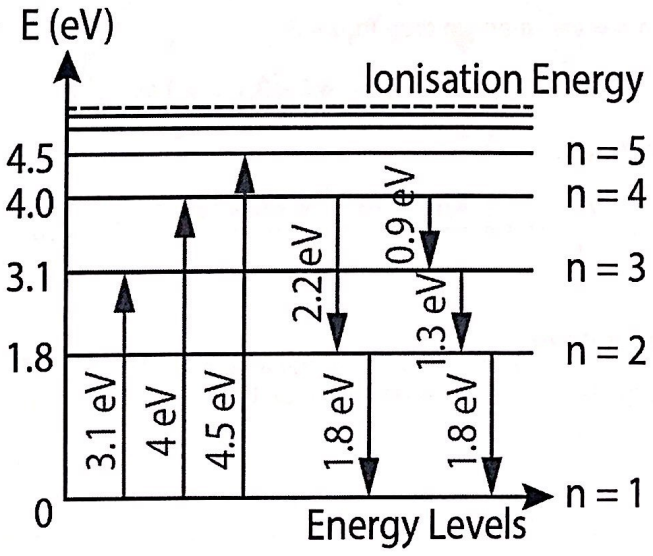

Figure 2.7: Example of an energy level diagram

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Example 2.3

Find the frequency of a photon absorbed that causes a jump to the energy level n = 3 for the atom in Figure 2.7 on page 53.

The energy change is given by:

$$E_p = E_2 - E_1 = 3.1 - 0 = 3.1\,\text{eV}$$

The frequency is given by:

$$E_p = hf \rightarrow f = \frac{3.1}{4.14 \times 10^{-15}} = 7.49 \times 10^{-14}\,\text{Hz}$$

An energy level diagram is used to show the energy levels for electrons in an atom in various excited states. Usually eV is used instead of J for ease of use.

From the ground state, $n = 1$, an atom can be raised to any higher energy state as long as the energy of the photon matches the change energy state. However, an atom can absorb *any* photon that has an energy greater than the ionisation value, as this is when the electron is 'ejected' from the atom and any excess energy from the photon is converted into kinetic energy (Photoelectric effect).

When an atom moves to a lower energy state, it does *not* have to drop straight back to its ground state. The energy of the photon emitted will be equal to the change in energy levels.

Example 2.4

An atom with energy levels give by Figure 2.7 on page 53 is initially in the 4th energy level. Find the photon with the largest wavelength that can be emitted.

The relation between wavelength and energy is given by

$$E_p = hf = \frac{hc}{\lambda}$$

So, for the largest wavelength, we have the smallest energy.

The smallest energy from $n = 4$is when we drop to $n = 3$,

$$E_p = E_4 - E_3 = 4.0 - 3.1 = 0.9\,\text{eV}$$

Now we can find the wavelength,

$$E_p = \frac{hc}{\lambda} \rightarrow \lambda = \frac{4.14 \times 10^{-15} \times 3.0 \times 10^{8}}{0.9} = 1.38 \times 10^{-6}\,\text{m}$$

2.3.3.3 Electron Standing Waves

Why do only certain energy levels exist? Again we look back at the dual nature of matter as electrons behave like both particles and waves!

Key Point :

An alternate model was proposed by de Broglie to overcome some issues with Bohr's model. As electrons have an associated de Broglie wavelength, stable energy levels are only those where electrons can form standing waves. De Broglie assumed these waves would be three dimensional, but they can also be shown on a two dimensional diagram.

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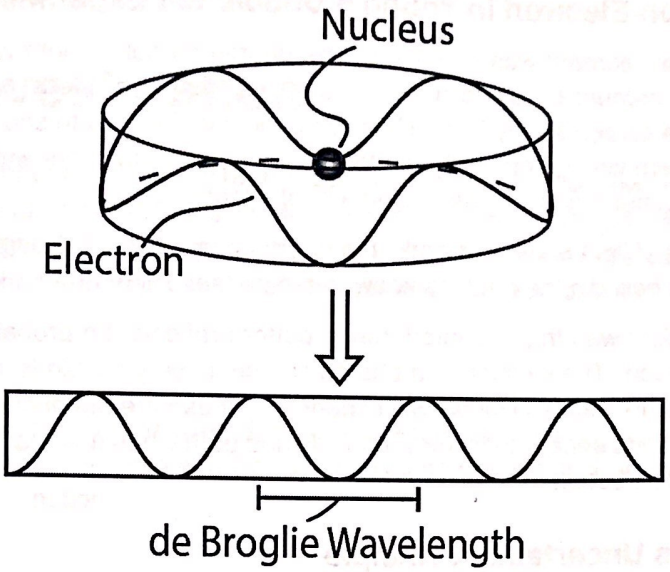

Figure 2.8: 3D illustration of standing electron waves

Figure 2.8 on page 55 shows us how these standing waves might look like in an atom. Notice how the de Broglie wave wraps all the way around the nucleus, with four periods. This would be for the $n = 4$ energy level. We could 'cut' the wave and lay it flat, which is also shown. The wave *perfectly* fits, so the circumference of the orbit is equal to four times the de Brogllie wavelength. This is why only certain energy levels are possible. At any other point, there is not an integer multiple of waves, for example 3.5λ fit into the circumference and the wave will *destructively interfere* with itself.

Example 2.5

Draw the standing waves for $n = 3$ and $n = 4$ of an atom.

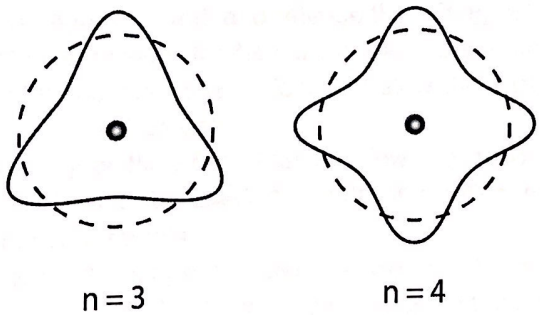

Figure 2.9: 2D illustration of standing electron waves

Figure 2.9 on page 55 shows how to draw these standing waves on paper, such as in a SAC or in the exam. Notice that for $n = 3$, there are 3 'peaks' and for $n = 4$ we have 4 evenly spaced 'peaks'. If it asked for say, $n = 5$, then there would be 5 peaks and the waves would fit *perfectly* into the circumference. This is all that is required to answer this sort of question. So, when we drop from an energy level of $n = 4$ to $n = 3$, the photon is emitted and the number of waves in the orbit drops from 4 to 3.

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2.3.4 Single Photon/Electron in Young's Double Slit Experiment

In 1909, an interesting experiment was contrived. The photon model for light was used to re-investigate Young's Double Slit Experiment by using a light source so weak that there would only be one photon between the slits and the screen at any time. The reason behind this was to see if light would now behave as a particle because there was no other photon for the interference to occur with, i.e. there was only one wave at any given time rather than two when using a bright light source.

The result in fact a series of light and dark bands, the exact same a normal Young's Double Slit Experiment. The key conundrum was how did the photons know to create these light and dark bands!

Essentially the 'explanation' was that the interference pattern reflects the probability of a photon taking a particular path to the screen. The location of a photon can be, in very simple terms, thought of these locations of high probabilities. In Young's Double Slit experiment, these are the bright bands in the interference pattern which is why photons appear to follow a preordained path when it actual fact they are materialising a governing probability distribution.

2.3.5 Heisenberg's Uncertainty Principle

While it would be very interesting to follow the path of a photon when it travels between the slits and the screen, this is quite impossible at our current technology levels. The reason is that the vast majority of our imaging devices rely on light (e.g. camera) or electrons (e.g. electron microscope) to produce an image. It would be like placing a number of tennis balls on the ground and asking a blind person using a cane or guide dog to determine the location of the tennis balls.

We cannot expect an accurate location as sweeping the cane side to side will knock the balls away from their position nor can we expect an accurate experiment because all of the balls have been moved! However, for large objects such as a building or a parked car, we could expect an accurate result because the there would be no effect of the object we're measuring. This is the Heisengberg's uncertainty principle.

Key Point :
In classical physics, studying an in-progress experiment is a simple matter. However, at very small scales, i.e. the quantum world, this is no longer an easy task. For example, it is not possible to measure both momentum and the position of a particle simultaneously.

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Area of Study 3

Practical Investigation

This SAC is a part of the course because it requires students to design *their own* practical investigation and analyse its results. It also allows then to test students on their *understanding* of Physics concepts. As such, a self-designed investigation that investigates the energy transformations of a balloon powered helicopter could potentially garner many more marks than analysing the motion of a tennis ball dropped from one metre above the ground.

3.1 Ideas

Optimally, investigations should aim to cover three or four Areas of Study to demonstrate *deep* understanding of the underlying Physics concepts and *how* they link together. As such, here is a short and obviously incomplete list of ideas for motion:

Hot Wheels Track:

1. Make a ramp and and vary the starting height. By varying the starting height, the initial gravitational energy changes. Analysis of how energy is conserved, transforming from gravitational potential energy to kinetic energy can be discussed as well as what 'energy loss' is. Or, since this is an inclined plane, Newton's Laws of Motion can be used to solve to acceleration, etc.
2. Make a loop-the-loop following the variable height ramp and find the minimum height at which a car will still make the loop-the-loop. Compare your results to a theoretical approach to see if the starting height is the exact same (if not, make your discussion about why this is so: energy loss, random errors, etc.)
3. Let the car run off the track and onto some frictional surface such as carpet. Find the distance the car travels on the carpet before coming to a stop and analyse the energy changes in your system.
4. Place a second car at the end of the track so that the first car collides with the second car sending it off onto the carpet. Investigate whether or not the distance travelled by the second car is equal to the distance of the first car in the previous part.

Trolleys, or other spring-loaded small cars:

1. Hold the spring loaded end against a wall and release the spring, allowing the cart to shoot forward. You can measure the distance travelled by the cart, or the average time taken for it to travel 3 metres (provided that the surface is frictionless and does not slow the cart down!). An analysis of spring energy to kinetic energy could then be done.
2. Use a table with one end higher than the other to allow the trolley to gently roll down the slope. Constant acceleration formulas can be used to find the theoretical time it takes to reach the end of the slope and compare it to your results.
3. Have the spring facing down the slope and place a wooden block at the end. When the trolley strikes the block, record the distance the trolley travels back up the slope and compare it to the initial distance. An interesting discussion of why the trolley does not return all the way up to the top of the slope can be made – energy losses, frictional forces, was the collision elastic or inelastic, etc.

Springs:

1. Attach weights to the end of a spring and measure how far the spring extends as you add weights. The force is given by the weights attached and, plotting against the extension, a rudimentary force-distance graph can be drawn. The spring constant can be found and a discussion on whether or not the spring is ideal can be made. Make sure not to overextend your spring!
2. After removing all the weights, add a small weight and allow the spring to oscillate, much like in the VCAA exam questions. Energy graphs can be drawn up, analysis of why the oscillation decreases over time (frictional forces) and calculations of speed/energy can all be included in your discussion!

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Again, these are just a few ideas to get you going – often the best pracs ones are where students come up with their own ideas and aim and carry it out (with *lots* of enthusiasm). Remember that these are just ideas for motion, you can also tackle fields or waves!

3.2 Typical Report Style

The key parts of a written report are listed below. The examples here refer to a practical about measuring the time taken for a ball to hit the ground after being dropped from various heights.

Aim: this outlines the *point* of your practical and should not be too long. Generally they begin with: "The aim is to investigate/analyse/explore..." and are open ended: "...the relationship/effect between the height a tennis ball was dropped and tht the time taken for tennis ball to hit the ground".

Hypothesis: this is your best guess as what will happen, and will be evaluated at the to see if it was right or wrong. Don't make it too simple such as: "The bigger the height, the longer it takes." State something *definite* such as: "The relationship between time and height is positive and linear." This opens the door for so much more discussion: do the results indicate it was linear? If yes, why? Relevant theories? If no, why not when comparing to established theory? What relationship is it? Was this expected when comparing to established theory? How does it compare to theoretical calculations?

Relevant or Established Theories: your teacher may or may not require this section. All it entails is you listing what theories already exist. In this case, it would be "Newton's Laws of Motion and the Constant Acceleration formulas".

Materials: if the practical exercise is fully provided for you, it is often acceptable to state "Refer to the attached practical exercise sheet". Just make sure to attach it!

However if you have used some other materials, make sure you are precise when describing them. The point of listing materials is to make your experiment *repeatable*. If you used standard tennis balls, state "standard tennis balls" so it's clear you didn't use novelty 3*cm* big tennis balls.

Method: again, if the practical exercise is fully provided for you, it is often acceptable to state "Refer to the attached practical exercise sheet". Just make sure to attach it!

Again, if you were told to make up your own method, be *explicit*. How many times did you repeat the experiment, what was the exact order, were you indoors or outdoors, if the latter did you wait for a calm day? This is all so someone else can come along and repeat your experiment and record similar results.

Results: this section is called 'results' for a reason, it should *only* contain your results data. Don't make any discussion here! However, do make an effort to make it neat by using tables. For example, you could have columns that state the trials: "Drop 1, Drop 2, Drop 3,...Average" and rows that state the heights: "1 metre, 2 metres, 3 metres,..." etc.

Analysis: this section is sometimes optional or you may be asked to combine it in with your discussion. The "Analysis" section is where you form your conclusions. For this practical, a plot of height against time would be ideal to show what kind of relationship exists. A line of best fit could also be drawn. Theoretical results can also be calculated and presented in a table with the practical results. A graph of practical results and theoretical results can also be useful for the discussion to see if the difference increases/decreases as the height increases, or if the two are always close.

If a calculation of error is required, this is the section for it. However, there are many methods for calculating error. Some teachers may only require a simple calculation of error: for example at $1m$ the average is $0.45s$, the maximum result was $0.48s$ and the minimum was $0.41s$. The largest difference is the error, $0.4s$, so the result would be written as $0.45 \pm 0.4s$. Longer, more accurate measures of error such as standard deviation may be required, however not only do these methods require longer calculations but also a larger number of trials.

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Discussion: this is where the majority of the marks are. A strong discussion should touch on the following points, *and* any others that your teacher has specified or are relevant to your specific practical.

1. **Summary of results:** this should cover how you executed the method, if there were any difficulties encountered following the method or any improvements in the method that could have been made. For example: "Although we successfully carried out the method, we encountered many difficulties with repeated trials as we were outdoors. The benefits of using a ladder outdoors would have been tremendous had the wind not interfered with our results. For future reference, an empty stairwell would be much better suited to this experiment as the weather would no longer cause variability in our results." Outline the major findings, but don't go into too much detail.
2. **Compare the findings with the initial hypothesis:** this was the reason for having a definitive hypothesis because we can how compare our guess to the results. If the hypothesis had just been: "A higher distance means a longer time", the discussion would go along the lines of: "And I knew it! Ha, 10 points for Gryffindor". With the complex hypothesis, "The relationship is linear", there is a much larger scope for discussion. Are the results actually parabolic? Would a linear line of best fit accurately demonstrate the relationship? Or, is a parabolic relation actually the best fit? It's okay if your hypothesis was wrong, some people would argue that an incorrect hypothesis is even better because it allows even *more* discussion. It also ties in nicely with the next part.
3. **How do the results and hypothesis stack up against the theoretical results:** here we begin by looking at the theoretical results that we obtained in the analysis section, and then compare them to the practical and hypothetical results. In this case we could refer to the fact that the relationship can be solved mathematically as $h = 5t^2$ using constant acceleration laws and g = 10. Then compare these to the real results: "Though we did get a parabolic relationship, it was not as steep as the prediction using theory..." and compare it to the hypothesis: "Clearly our hypothesis was incorrect as it is disproved by the results and the theoretical calculations in the analysis section". Finally, outline any *limitations* of the theory or any assumptions that were made that shouldn't. For example: "However, the theoretical results do not include the very real presence of air resistance. If air resistance was included then ... Moreover, the value of 10 was used for the gravitational field strength which is far from realistic ..."
4. **Errors:** are they present in your answers, what type are they and how accurate actually are your results? Even if your teacher did not require you to do an analysis of the errors and/or quantify them, you should *always* talk about errors. There are three main types of errors: random errors, systematic errors and human errors. Random errors are always present in experimentation and as such cannot be directly eliminated. The only way to reduce their effect to to increase the number of trials, averaging out fluctuations. Systematic errors occur when an error is made once and constantly repeated. For example, if the one metre ruler used was actually 1.2m long and no one noticed, then all the trials would have been consistently done at the wrong height leading to wrong results. The main method of reducing systematic errors is to continually check against each step with the method provided and check all equipment. Human error occurs when the presence of humans causes the error and the results go awry. For example, when dropping the ball the dropper could have unintentionally been throwing the ball towards the ground. Had a machine or a robot been used, this error would have been avoided. Or, another example is recording data. Perhaps a number was written down or copied incorrectly. Also mention the steps taken to avoid or reduce errors, such as repeated trials and double checking results were accurately recorded.

Conclusion: this should be clear and concise, beginning with a reiteration of the aim, for example: "We successfully investigated the relationship between the time taken for a tennis ball to hit the ground and the height it was dropped at". This is followed by about one line for each point in the discussion, i.e. "We found that from our results the relationship was parabolic... This contradicted our hypothesis of a linear relationship... there were errors in our results that were most likely introduced by..." etc. Overall, the point of this prac is to ensure you understand the concepts behind the results. Make sure to constantly refer to these Physics concepts and formulae!

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Part III

Final Exam

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Section 1

Tackling the Exam

In any one Physics exam, there are three typical question types. Listed below are each general types and my suggested approach to answering each one.

1.1 Question Types

1.1.1 Short answer questions

These questions are aimed at testing your knowledge of Physics. They aren't designed to trip you up which is why they are only worth one or two marks. Figure 1.1 on page 61 shows a one mark question and a two mark question.

For any one mark question, you either get one or zero as there are no half marks in the exam. So, it might seem reasonable to just write the answer and not do any working, but this is not my approach. I prefer to write one line of reasoning or calculation so I make sure that my reasoning is not flawed. Moreover, when I go back to double check, that one extra line of working makes it much easier for me to check my reasoning was correct.

For the two mark question, working is a must. Not only are there potential marks for working, but if the final answer is incorrect I can still salvage one mark for my working out of two which is a lot better than nothing. When writing my working, I prefer to state the formulas I use before substituting and evaluating. However, stating the formula is not mandatory as the mark(s) for working are awarded to an equation with correctly substituted numbers.

Question 14 (3 marks)

An ideal transformer has 130 turns in the primary coil and 5200 turns in the secondary coil.

a. A DC voltage signal of 12 V is connected to the primary coil.

State the value of the steady voltage at the output of the secondary coil. 1 mark

DC in → No change in current → No flux → No output Voltage

0 V

b. An AC voltage signal is then connected to the primary coil and an AC RMS voltage of 400 V appears at the output of the secondary coil.

Calculate the **peak** value of the voltage at the input of the primary coil. 2 marks

$V_S = 400V \quad \frac{V_P}{V_S} = \frac{N_P}{N_S} \rightarrow \frac{V_P}{400} = \frac{130}{5200} \rightarrow V_P = 10V$

$V_{peak} = \sqrt{2}\, V_{RMS} \rightarrow V_{P,peak} = \sqrt{2} \times 10 = 14.14\,V$

14.14 V

Figure 1.1: Question 14 from the 2014 VCAA exam with sample solution

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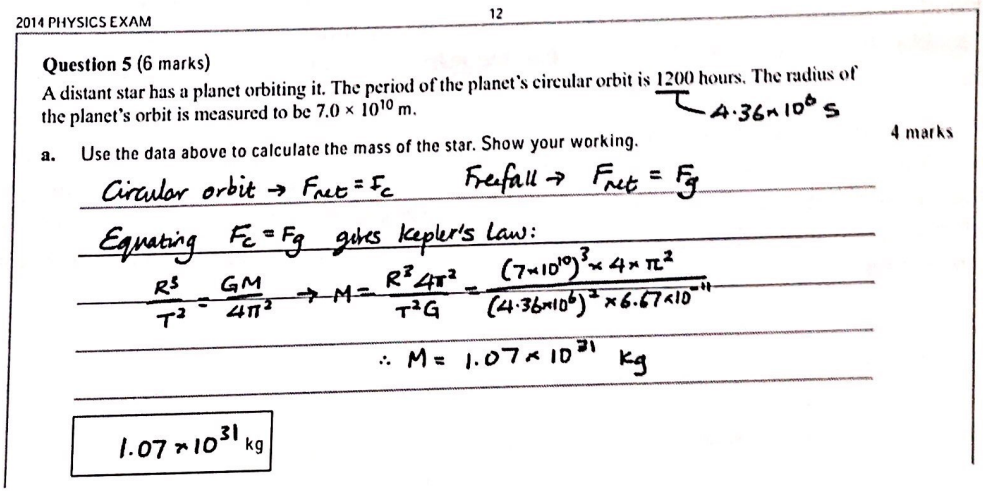
1.1.2 Long answer calculation questions

In my mind, there are two general "classes" of long answer questions. The first "class" involves difficult calculations that must be done carefully while the second "class" involves difficult concepts that may be explained in your answer.

Figure 1.2 on page 62 shows a typical long answer (three or four marks) question for gravity and satellites. The reason I have given so much working is twofold: if my final answer is wrong I can still garner some marks for working and in my working I have shown "good physics". Sometimes, it is not enough to have lines of calculations, if there is no apparent *reason* for them. On past assessment reports, it has been stated that marks should be awarded for "good physics".

To me, "good physics" means showing an understanding of the underlying concepts and assumptions made rather than just plugging numbers into a formula from my cheat sheet. So, for question 5a of the 2014 VCAA exam I clearly state my assumptions that the orbit is circular and the net force is due to free-fall. Then, I go into the relevant calculations.



2014 PHYSICS EXAM 12

Question 5 (6 marks)

A distant star has a planet orbiting it. The period of the planet's circular orbit is 1200 hours. The radius of the planet's orbit is measured to be 7.0×10^{10} m.

4.36×10^6 s

a. Use the data above to calculate the mass of the star. Show your working. 4 marks

Circular orbit → $F_{net} = F_c$ Freefall → $F_{net} = F_g$

Equating $F_c = F_g$ gives Kepler's law:

$\frac{R^3}{T^2} = \frac{GM}{4\pi^2} \rightarrow M = \frac{R^3 4\pi^2}{T^2 G} = \frac{(7\times10^{10})^3 \times 4 \times \pi^2}{(4.36\times10^6)^2 \times 6.67\times10^{-11}}$

$\therefore M = 1.07 \times 10^{31}$ kg

1.07×10^{31} kg

Figure 1.2: Question 5a from the 2014 VCAA exam with sample solution

1.1.3 Explanation questions

The key to exam explain-type questions is to ask yourself "Why?". You need to keep on asking "Why?" until you reach an underlying Physics concept or formula that has already been established. So, for question 5b in Figure 1.3 on page 63 I begin by stating the answer that it is not possible to determine the mass. Then, I have to ask myself "why?". Well, from 5a I saw that *m* did not appear on either side of the equation. Why is this so? It's because I used Kepler's law. Where did this come from? It came from equating the centripetal force and the force due to gravitational attraction. These are formulas are already well established and would be the end of my answer

So, it isn't just enough to say that it is not possible and the reason is *m* does not appear in the formula. To definitely get two out of two, you want to keep going until you hit a key formula or concept in your answer to, again, show "good physics" rather than application of formulas from your cheat sheet.

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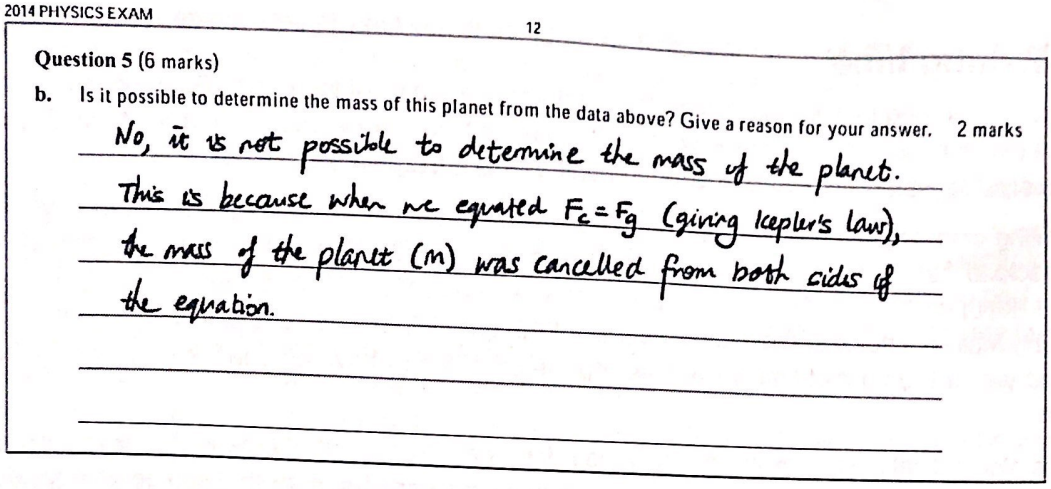

2014 PHYSICS EXAM 12

Question 5 (6 marks)

b. Is it possible to determine the mass of this planet from the data above? Give a reason for your answer. 2 marks

No, it is not possible to determine the mass of the planet. This is because when we equated $F_c = F_g$ (giving Kepler's law), the mass of the planet (m) was cancelled from both sides of the equation.

Figure 1.3: Question 5b from the 2014 VCAA exam with sample solution

1.2 Reading time

When I was preparing for my exams, reading time somehow was simultaneously the most overrated part of the exam and the most underrated part of the exam. It was underrated because quite a few people would just read through the exam in 5 minutes, then daydream until writing time started. It was overrated because of the "importance" people would often put on it – that it was your one chance to go through the paper before, or it was a chance to decide what order you do the questions, or even a chance do some of the multiple choice questions in your head.

But for me, reading time was just a chance to make sure that everything I expected to be in the exam was there, and there weren't any major surprises. It was a chance to sit down and clear my head before I put my pen to paper. It was my chance to focus and mentally prepare myself. That was my tactic.

However, that was just *my* tactic. Everyone has their own tactics and plans for what reading time will be like. You need to figure out which tactics you prefer and then use them. The common ones that you can choose from are:

1. Read through the exam (duh).
2. Re-read the exam to make sure you understand all the questions.
3. Decide what order to the questions in.
4. Try to answer some of the questions.
5. Try to eliminate some of the options for multiple choice questions.
6. Take a few minutes to focus.
7. Get your brain in 'exam mode'.
8. Check your cheat sheet as you read to make sure everything is on there.
9. Look up any words that you don't know in a dictionary.
10. Read the formula sheet to familiarise yourself with it.
11. Read all of the instruction boxes.
12. Drink some water to keep yourself hydrated.
13. Check your stationery.
14. Make sure you have a good view of the clock.

In my mind, it is possible to shorten this list (which is not exhaustive) by preparing *effectively*. For example, you can go through the formula sheet from past exams to familiarise yourself with it. You can read all of the instruction boxes from previous exams as these hardly change. You can make sure you can see the clock and have a drink as you sit down before reading time starts.

But, as I've said, everyone does have their own tactic. It is up to you to form your own tactic, and I do hope that I've given you some ideas on what you will do. I also hope that you practise reading time before the actual exam by taking the 15 minutes before each of practice exams you do to get used to reading time!

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1.3 Writing time

It goes without saying that writing time is *the* most important part of the exam. But it doesn't mean is that you can't take a minute to drink some water or to use the toilet. While you need to get through your exam, you also need to make sure you look after yourself as it is a long exam.

Now, writing time begins right after reading time which means the exam should be fresh in your mind. If you've decided that you will spend reading time deciding what order to do the questions in, then even better! For me, I like to go through chronologically through the core sections then the detailed study. My reason for leaving the multiple choice section to last is that if I run out of time, I can still guess the questions I haven't answered yet. If I run out of time while finishing the core, then I have no chance of guessing the correct answers.

Timing is also critical in your exam, because you don't want to run out of time in the exam. It's a horrible feeling having to leave an exam knowing that you could have done better or that you could have done such and such questions but you just ran out of time. My tip is to either come in prepared with a timing plan, or make an ad hoc one.

- **Pre-planned method:** before you go into the exam, you find the sample coversheet for the exam that VCAA posts every year. This will show you mark break down. You can then decide how long you want to spend on each section and write out the starting and ending times on your cheat sheet. Alternatively, you can judge from practice exams you've done: if you find that on average you can finish motion in, say, 30 minutes, electronics and photonics in twenty minutes etc then you can use those times to base your plan on.
  In the exam, you would then follow your planned times to make sure you don't go over time. If you realise one question/section is moving too long, then move on to the next one and come back to it at the end!
- **Ad hoc method:** for each question, there is an allocated total mark. I'm not taking about Q1a., Q1b., but rather the total amount for question 1. For the 2014 exam, you start at 9.15 am and question 1 is worth 9 marks. So if you want to spend a minute per make, you would write "9.15 start –> 9.24 finish" next to question 1.
  If you finish early, say 9.20 am then for question two (worth 11 marks) you would write "9.20 start –> 9.31 finish (+ 4 min)". The "+4 min" indicates that you are running 4 minutes ahead of time, so you have an extra 4 minutes at the end of the exam to double check.
  If you're running late on question one, you can either decide to skip it till later or go over time with it. If you do and it takes you till, say, 9.26 then on the start of question 2 you would write "9.26 start –> 9.37 finish (-2 min)". The "-2 min" tells you that you're running 2 minutes behind and you need to make up for it somewhere else.

Obviously, the ad hoc method takes much longer to do during an exam. It also means that you have to do a running total: for example if you're 4 minutes in front in question 1, 5 minutes behind in question 2, 3 minutes in front in question 3, then your running total is "+2 min". The upside is that you always know if you're running under or over time.

One other issue with the ad hoc method is what if you want to have some time at the end to double check? Then it's no longer a mark a minute. Rather, if you wanted 15 minutes at the end to double check it would be a mark every 54 seconds ... which means that to calculate the amount of time for 9 marks you would have to do $9 \times (54 \div 60) = 8$ min 6 sec... you can see how impractical this becomes if you do it for every question! So, if you just come in prepared with a time plan that has been adjusted for double check time, then it's a *lot* easier.

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Okay, so you have a general idea of what you're going to do, how long you're going to spend doing it and when to move on if it takes too long, but how about actually doing "it", the questions!

My general guide goes like this:

1. Skim read the question and all of its parts to get a good idea of what it's asking. If you still remember the question from reading time, that's great because you can skip this step!
   This is just to make sure that you don't go and solve for the kinetic energy *and* velocity in part (a) when in fact it asked you to only find the kinetic energy in part (a) and the velocity in part (b)... I've been guilty of this plenty of times and I've had to draw arrows every which way for every question to show what working belonged to (a) and which belonged to (b).
2. Go back and highlight/underline any key words that you will need.
   Questions appear in blocks, so you don't want to get up to part (d) and start losing track of where all the data in the question is.
3. Draw a diagram if you think it would help.
   Usually VCAA provides all the diagrams you need, but if you think another diagram (e.g. free body diagram) or graph would help you then definitely draw it. If it's for an explain-type question then all the better!
4. Show full working, taking care to make sure it is logical.
   If you don't manage to get the right answer, you definitely want those consequential marks. Look back to the sample solutions for various question types to get an idea of how much detail you should put for your working. Show the formulae your are using with substituted values as there is a mark for showing a formula with substituted values and a correct answer.
   Only write in the borders because your exam will be electronically scanned.
5. Write out your final answer and indicate that it is your final answer.
   This can be as simple as writing your answer in a box. At worse, you will write a sentence a few words long, e.g. "So, the velocity is 5m/s". That clearly indicates your final answer to your examiner.

Now, that's all well and good and I hope that you stick to my general guide! But what happens if you get stuck on a hard question?

1. Don't panic. If you've done your revision and you're finding a question in the exam hard, chances are the rest of the state will be feeling the same way.
2. Look back to the key information you highlighted/underlined. Try to find a link between them.
3. Check the formula sheet or your own cheat sheet to see if there are any relations that involve the information in the question.
4. If the space for the answer is small, you might just be over thinking it – try to look for a simple solution. On the other hand, if the space is large, prepare yourself for a lot of work.
5. If the question is a multiple choice response, try to eliminate some of the options. For example, look at the units, or substitute the answer into the question to see if it works out.
6. Worse case scenario: You still have nothing and looking at the amount of time you've already spent on the question you realise you need to move on:
   (a) Write some physics formulas or concepts that are *relevant* to the question. Don't over do this, it's just to show that you understand physics, just not this particular question. A cynical way to think of it would be you're after pity marks.
   (b) Put a star or some kind of mark next to the question so you know to come back to it. If it's a multiple choice question, guess an answer and put a mark next to the question number. This way, if you don't have time to finish the question at the end, you still have a guess on your response sheet.
   (c) Move on. It's the only thing you can do. Move on and get the rest done so you can come back and nut out the hard question.

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1.4 What to do before your exam ends

Double check everything! When I was going through Physics, I was the kind of person that would get all the trick questions right but get several "easy" ones wrong. Why? Because I'd rush and not read the question properly if it seemed easy. Eventually, I managed to cut down on my mistakes by taking care with each question and allocating time at the end specifically for double checking. When I double check, these are the common things I do:

- Make sure I answered the question in the right units.
- Make sure I wrote my answer in the box provided.
- Make sure I've answered the right question. If the question asks for kinetic energy, then I make sure I solved for kinetic energy and not for, say, the velocity instead.
- If I couldn't fit all my working/explanation on the page, draw an arrow or make some mark for the examiner to show where it continues.
- Double check that my answer is reasonable. If I find the energy of a photon is 680 eV, then I know I've made a mistake.
- For long calculations, re-enter it into my calculator to make sure that I didn't type it in wrong.
- For multiple choice questions, make sure that I've shaded the right box on my response sheet.
- Make sure that I answered (or tried to answer) every question.
- Make sure I filled in my student number correctly.
- Make sure I filled in the multiple choice response sheet.

Now, some of you will notice that you can, and probably should, do some of these after each question you finish. If you double check that you answered the right question, make sure that your answer is reasonable, in the right units and in the provided box, then that is an excellent habit to get into.

1.5 How to use assessment reports

Assessment reports are like gold. I can't stress enough the importance of looking through the assessment report for every past VCAA exam you do as it changes every year. There are two main sections to assessment reports; the general comments and the specific information. The first section can seem a bit daunting because there is a *lot* of writing but it is full of useful information, for example the 2014 assessment report states things such as:

- The answer space given and the number of marks allocated to a question should be used a guide as to the amount of detail required in an answer.
- Attempting a question a number of different ways will not be awarded any marks unless all methods are correct. Students are advised to neatly cross out any working they do not want assessed.
- Students should be encouraged to set out their work clearly so assessors can follow what they have done. In questions that involve a number of steps, it is helpful if the student explains all their working.
- In questions that require explanations, students should carefully consider what the question is asking and answer accordingly. They should not simply copy information from their sheet(s) of notes, as this can result in the inclusion of irrelevant, contradictory or incorrect material.
- When responding to questions that required an explanation, many students answered in dot-point format. This may help to ensure good, concise answers. There is no need to restate the question in an answer.
- The use of equations or diagrams in questions that require an explanation can sometimes assist. It is important that diagrams are sufficiently large and clearly labelled. Graphs and sketches should be drawn with care.
- Students' attention should be drawn to the instructions for Section A, 'In questions worth more than 1 mark, appropriate working should be shown'. Full marks may not be awarded where only the answer is shown, and some credit can often be given for working even if the final answer is incorrect.
- Students are also reminded of the instruction for Section A, 'Where an answer box has a unit printed in it, give your answer in that unit'. Students should be discouraged from changing the unit.

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- It is important that students show the numbers substituted into formulas/equations. The formula alone is generally not worth any marks.
- It is expected that formulas be copied accurately from the formula sheet provided with the examination or from the student's sheet(s) of notes. Derived formulas from the student's sheet(s) of notes may be used. However, they must be correct and appropriate for the question.
- Students need to be familiar with the operation of the scientific calculator they will use in the exam. Calculations involving powers of ten sometimes caused difficulties for students. Students must ensure that the calculator is in scientific mode and that it does not truncate answers after one or two decimal places.
- The rounding-off of calculations should be done only at the end, not progressively after each step.
- Answers should be simplified to decimal form.
- Where values of constants are provided in the stem of the question or on the formula sheet, students are expected to use the number of significant figures given.
- Care needs to be taken when reading the scales on the axes of graphs.
- Arrows representing vector quantities should be drawn so that they originate from the point of application. Where appropriate, the length of the arrows should indicate the relative magnitudes.
- Students should ensure that their answers are realistic. Illogical answers should prompt students to check their working.

Like I said, the general comments are a veritable gold mine of information about *how* to answer questions. I've given you examples on how to answer various questions types such as long answer and short answer, but these dot points will be what your VCAA exam marker will be following. I also find, and this comment is not directed at anyone in particular, that the general comments answer a lot of frequently asked questions such as: can I change the units in the box? can I use dot points? do I need to show working?... they're all answered in the assessment report!

Another common question is: what's the hardest thing in Physics. Again, I can give you my personal opinion, but the answer in the assessment report is much better because it is based on how the entire cohort went in 2014, rather than the ramblings of one past VCE student!

Areas requiring improvement included:

- connected bodies
- energy conversion and conservation in springs
- complex projectile motion
- apparent weightlessness
- modulation
- explaining direction of induced current using Lenz's law
- operation of transformers and how they work in a power transmission system
- understanding of series circuits
- explaining aspects of the photoelectric effect
- applying the concept of path difference in interference patterns
- explaining electron and X-ray diffraction patterns
- electron energy level diagrams and associated emissions and absorptions
- how the wave nature of matter can explain the electron energy levels.

These lists are also updated every year so if you look back through a few past exams, you'll definitely get a sense of what questions are hard/not well done. Some examples would be "energy conversion and conservation in springs" which has cropped up almost every other year for the past decade and "explanation of induced current using Lenz's Law" which has gotten beefed up in the past few years. The main caveat, however, is that if you spent too much time looking back through past VCAA reports, you might get caught in the trap of trying to guess what VCAA is going to do next for its exam... you want to spend your time getting ready for the exam not becoming a fortune teller for the exam questions! A general idea of past exams is good, just don't start trying to find patterns and make predictions.

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The specific information is great because it is, well, very specific. So, for example you go through the 2014 exam for revision and you had now idea how do to question 2 of the motion section. Well, by looking at the assessment report you can actually see how other students faired and get an idea of where you lie.

Question 2d.

Marks	0	1	2	3	4	Average
%	74	15	3	1	7	0.5

Average of 0.5 out of 4... ouch.

Another way to use the specific information is to *deliberately*, yes deliberately, go looking for hard questions to practise on. To me, this is a very good idea for those of you who are aiming high because commercial (non-VCAA) exams typically do not have statistics on how other students fared. So, by hunting through assessment reports for hard questions, you're making the most of your resources. Personally I would classify any question where over 50% of the state earned 0 marks a difficult question, but your standards might be higher or lower.

The assessment reports do not, however, give you fully worked solutions. For fully worked solutions, I would try VicPhysics or iTute if you're struggling to follow an assessment report as they do skimp on a lot of the working. The thing they do offer, however, are comments on the common errors that students made. You might find this interesting, or you might not find it interesting, but personally I make far too many silly mistakes so being aware of common errors meant I knew where to take care.

Finally, the detailed study section is *extremely* sparse - for the 2014 detailed study, there was only one comment for the entire area of study for sound. Again, if you are looking for worked solutions you will have to check out vicphysics or itute as they do publish fully worked solutions for VCAA exams every year.

1.6 What exam mark do I need for...

Don't. Don't do this to yourself. Just don't. There are plenty of "guides" on how to predict your study score and the majority of them are based on normal distributions. I won't go into any of the methods because it doesn't really matter. All you really need to keep in mind is that your final study scores are normally distributed, which means everything is *scaled.* If the exam is easy one year, that *doesn't* mean that everyone gets high scores. The same proportional of students will get 30+, the same proportion will get 40+, etc. It just means that the requirements for getting a C, C+, B, etc. are changed. These are referred to as 'cut-offs'.

So, if the exam is easy one year, it might have a high cut-off in order to get an A. If the exam in another year is hard, the cut-off in order to get an A might be lower. In this way, the distribution of study scores is the same regardless of the difficulty of the exam. In other words, 140/150 on one exam is not the same as 140/150 on a different one. It all depends on the grade distribution of the exam. So, if you want to get a specific study score, then don't concentrate on what exam mark you will need to get. In fact, it would be better if you spend your time revising rather than worrying what mark you need. But, if you must, then I would suggest you aim at letter grades like B+, A, A+ etc. if you want a target.

1.7 What can I do right now?

There are heaps of things that you can do. You don't have to be in SWOTVAC to start studying for your exam! My first piece of advice would be to get your cheat sheet together. More than likely you've finished all of your core studies and you've had all of your major SACs. At most schools, you're allowed to take a cheat sheet into your SACs. So, if you still have your past cheat sheets it will significantly reduce the amount of time needed to make one.

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So why make your cheat sheet so early?

- It's much better to do it now than leave it to the night before your exam. You'll be able to carefully craft it yourself instead of rushing to just get everything onto the one page.
- If you get your cheat sheet done early, you can practise using it in your trial exams. This way, you'll become more familiar with your cheat sheet.
- There's still time to edit or change your cheat sheet. If you're going through a practice exam and you suddenly you realise you've forgotten a crucial definition, you still have time to add it in. Imagine if you only realised you forgot that definition during the exam.

Okay, so hopefully I've convinced you to make a start on your cheat sheet. So, what should you put on your cheat sheet? I think there are two general approaches:

- **"Shotgun" spread:** this is where you try to get everything and anything on your cheat sheet. For this method, I think it would be best to start with a pre-written commercial cheat sheet and then add your *own* comments to it. Adding your own comments is important because no commercial cheat sheet will have exactly everything you want. It also helps immensely if you finish your cheat sheet before you start practice exams because you'll realise what formulas/definitions are missing on the commercial cheat sheet.
  For an idea of where to get commercial cheat sheets, try looking at iTute unit 3 and unit 4 cheat sheets or look at ones posted by past students on the ATAR Notes Forum.
- **"Sniper" precision:** this is where you take a very personal approach and make a cheat sheet that is tailored to your own needs. For example if you find that you are very confident with motion, but struggle with definitions for light and matter, then a general commercial cheat sheet might not work for you. A personal cheat sheet that has just key formulas for motion but lots of past exam questions, definitions and examples for light and matter would be much better suited to you.
  The obvious issue with this method is that you need to know your strengths and weaknesses before you start making a cheat sheet. An easy way around this issue is to just base it off your SAC scores for each area of study – the lower you score the more you put on your cheat sheet for that area of of study. Then, as you start doing practice exams you can further tweak your cheat sheet as you learn more about your strengths and weaknesses.

Okay, so cheat sheet: tick. What next?

There's still *plenty*. To start off with, you can start doing past Unit 3 exams. I suggest doing individual Unit 3 exams, rather than starting off with a Unit 3 and 4 combined exam or doing (separate) Unit 3 and Unit 4 exams together one after the other for a number of reasons:

- For starters, unless you've sat a 2.5 hour exam before, most likely you won't be used to sitting down and working through an exam. By starting with a shorter 1.5 hour Unit 3 exam, you can start working up to the longer ones like Physics post-2013.
- If you combine two separate exams, then that's even an even longer exam (1.5 + 1.5 = 3 hours). Again, start with something shorter and then work up to your final exam.
- It's probably been a long time since you looked at Unit 3. If you try do a combined exam all in one go, you might be put off with your results. So, if you single out Unit 3 you'll be specifically revising Unit 3.

Now, after you've done a couple Unit 3 exams, you'll get a good idea on what you should be revising. Again, looking back at high yield vs. low yield you'll realise I'm suggesting doing a lot of past exams that are high-yield. What about reading the textbook or doing textbook exercises? What about that one exercise that you promised yourself you would do when you got around to it? Forget it for now. Concentrate on getting used to doing exams, utilising reading time and drafting up a cheat sheet. If, and only if, you find you're struggling with a particular section, then go back and revise using your textbook or ask your teacher. Otherwise, exam questions are much more important than textbook questions.

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1.8 What should I do between now and then?

Practice exams. Seriously, I cannot stress this enough. You need to do practice exams.

And not just any old "practice" exam, but a well-written practice exams with solutions that you can do under timed conditions. If you're going along and you find that, say, commercial exam A is really easy then you should consider *changing* to VCAA or other commercial exams. You need a bit of challenge to give you some pressure when you do a practice exam. That said, be realistic. If you're going after exams that are *much* harder than VCAA exams, forget about it. You're probably set for Physics if you're finding VCAA exams a breeze and you need to make sure you feel the same way about all your other subjects.

The process I went through when doing practice exams was something like this:

1. Find a quiet place where I would *not* be disturbed. I liked to do practice exams in my bedroom.
2. Make sure I have access to a clock and make a quick note of how long I want to spend on each section.
3. Decide if I want to use my cheat sheet or not.
   (a) If I want to practise my speed, then I won't use a cheat sheet. This will force me to do questions when I see then, rather than spending time checking my cheat sheet. By practising this way, by the time I get to the exam I should be fast and not need to constantly rely on my cheat sheet.
   (b) If I want to practise getting as many marks as possible, then I will use a cheat sheet. This is because I will be using it to make sure I'm using the right formulas, definitions, etc. when I'm working through the exam, and when I finish my exam I will use my cheat sheet for double checking.
4. Practise my 15 minutes of reading time. Seriously don't skip this.
5. Do the exam under timed conditions. I will make sure that I do not go over time and I keep the pressure on myself to do the exam well and do the exam fast.
   I find that doing exams under timed conditions is especially important because it gives a very different feeling. You want to practise the same way your actual exam will be. There is a HUGE difference between doing an exam in silence under timed conditions while a VCAA invigilator walks around the room and doing a practice exam at the kitchen table with all my notes and the smell of dinner cooking in the background.
6. I finish the exam.
   This might sound silly, but you seriously want to finish the exam all in one go. What you don't want to do is spread the one exam over a few days or even a week. If you're going to do that you might be better off doing Checkpoints or other exam-style questions. This is because you won't get the full exam "experience" with things like fatigue, getting hungry or having to rush off to the toilet when you only have 30 minutes left and you're halfway through a question. Try not to "stop the clock" for any trivial reasons.
7. Take a break if I need it.
   Having just finished a 1.5 hour (Unit 3 or Unit 4) or a 2.5 hour (Combined Units 3&4) or a 3 hour (two separate Unit 3 and Unit 4) exam, often you will need a break. Take it. If you don't, you won't be able to properly look through your exam when marking it.
8. Mark my exam and take a note of what I got wrong.
   The reason for marking the exam is obvious, but what's equally important is making sure you know what you got wrong and *why*. Was it just a substitution error? Calculation error? Or was there something that you just simply did not understand? By making note of what you get wrong you can quickly see if there is a pattern that forms. Excel is very handy for keeping track of how you go in your exams and looking at which questions you get wrong.

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Below is just a sample table that you might find handy for keeping track of how you go. You can quickly make one in Excel for yourself!

	Trial Paper 1	**Trial Paper 2**	**Trial Paper 3**	**...**
Date:	19/9/2015			
Mark (%):	80%			
Letter Grade (if VCAA):	A			
Marks lost:	30			
Motion	10			
Electronics & Photonics	5			
Electric Power	9			
Light & Matter	4			
Detailed Study	2			
Why I lost marks:				
Didn't understand the Q				
Read the Q wrong	x			
My explanation was wrong				
Didn't know what formula to use				
Substitution error	x			
Power of ten error				
Wrong units	x			
Calculator Error	x			
(Other) Silly error	x			
Ran out of time				

I definitely recommend that you personalise your own table. For example, you could break down "Motion" into separate topics. This way you can see exactly where you're losing marks: is it all areas of motion, or are you just losing marks on circular motion? As you add to your table, you want to continually consider:

- Was everything I needed on my cheat sheet? Do I need to change anything?
- Am I continually getting one type of question wrong, e.g. explain questions?
- Is there a topic that I always seem to get wrong, e.g. Lenz's Law?

You may have a different plan for practice exams, and that's great! Just make sure that you keep on adjusting your cheat sheet so you have a "perfectly tuned" one for the final exam and help keep a track of which questions you get wrong so you know which topics you need to improve on by going back to the textbook or asking your teacher.

I wish you all the best for Physics and I hope that I cleared up some of your questions as well as giving some insight into VCE Physics exams.

Good Luck!

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ATARNotes

Physics Units 3&4

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"the notes provided a concise overview of the course content, and going through them is a much faster and more efficient way of revising as opposed to rereading the textbook... it is almost as though a real person is explaining it to you – complex ideas are explained in simple terms, and this makes it much more inviting to read through."

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